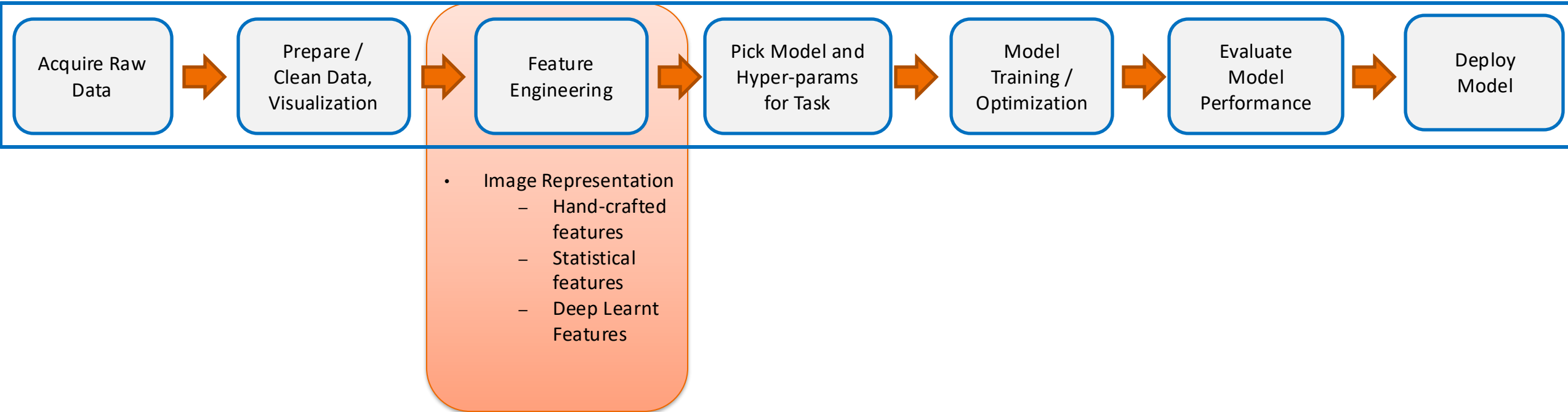
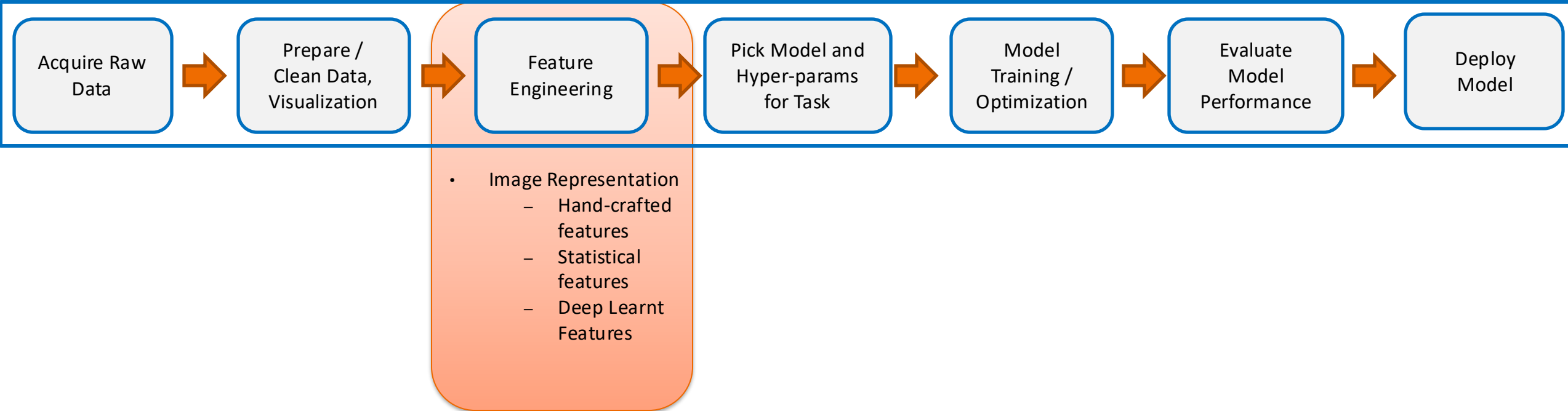
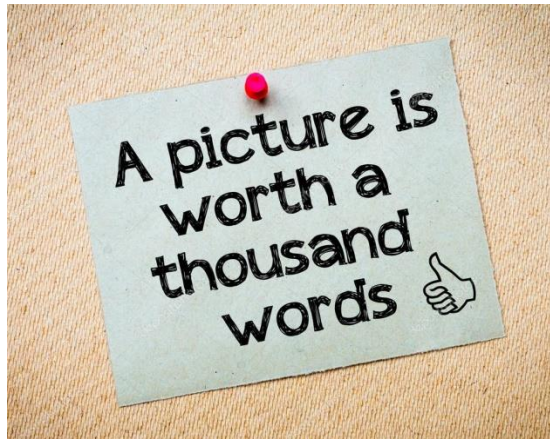


Focus for this lecture





AI and Problem of Perception





Why is it challenging?



Occlusions/Truncations

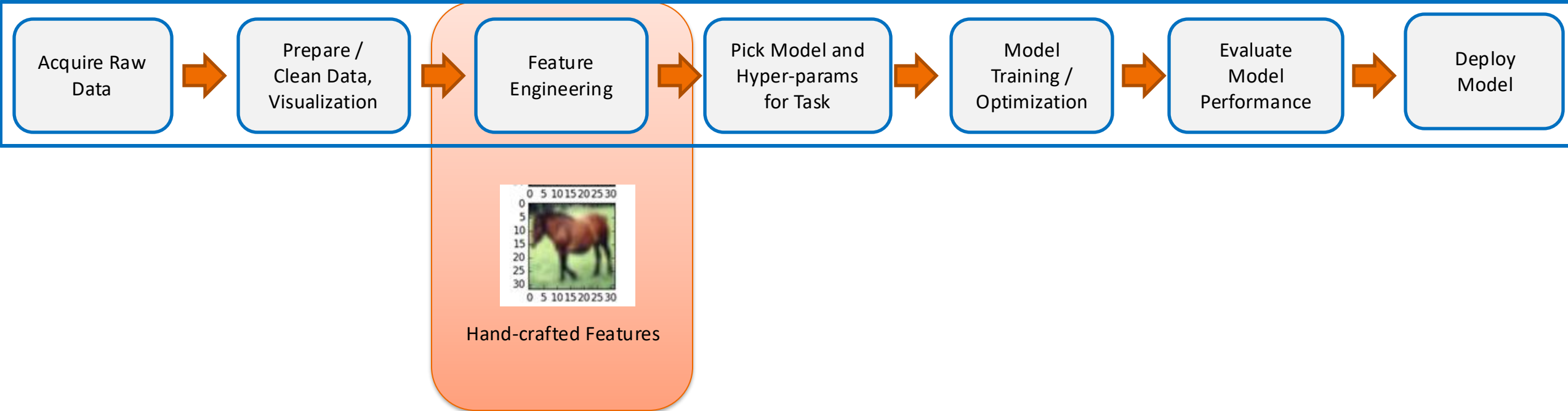
View Point Variation



Intra class variations



Inter class variations



Hand-crafted features

What is a digital image?

- 2D matrix of intensities (gray or color values) or numbers

100	50	0	150
90	255	70	70
200	150	255	50
0	100	80	0

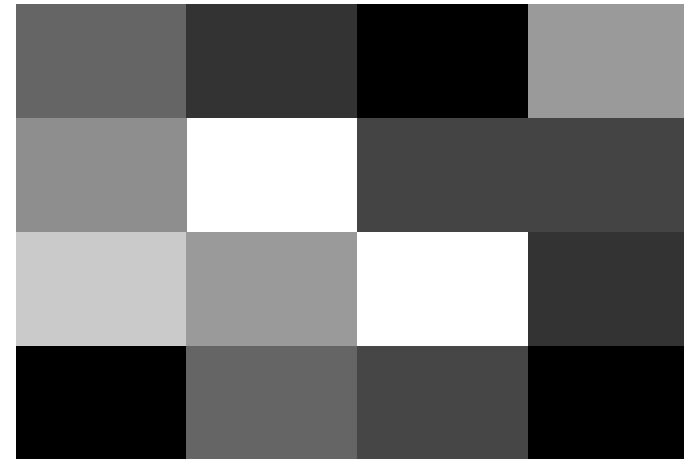
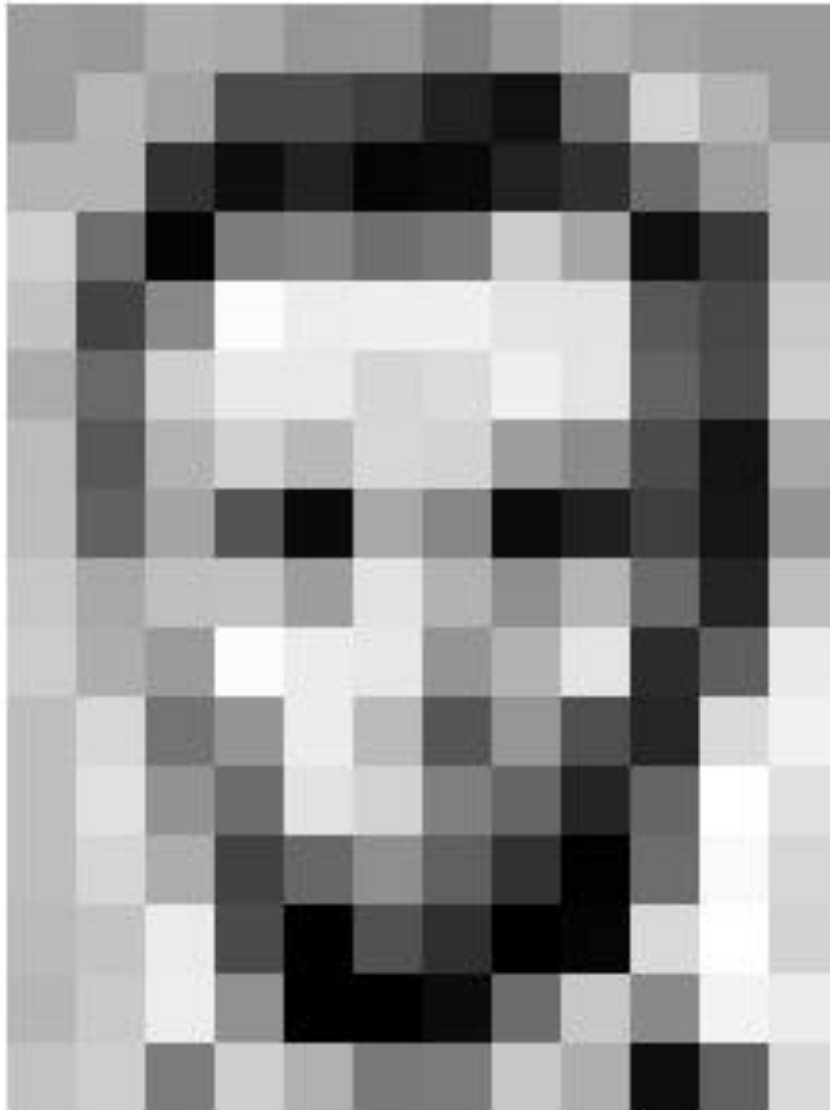
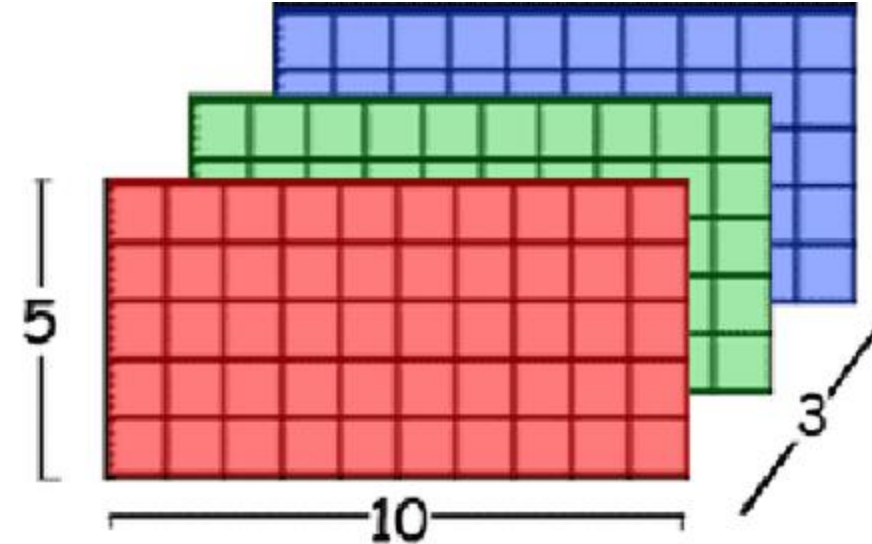
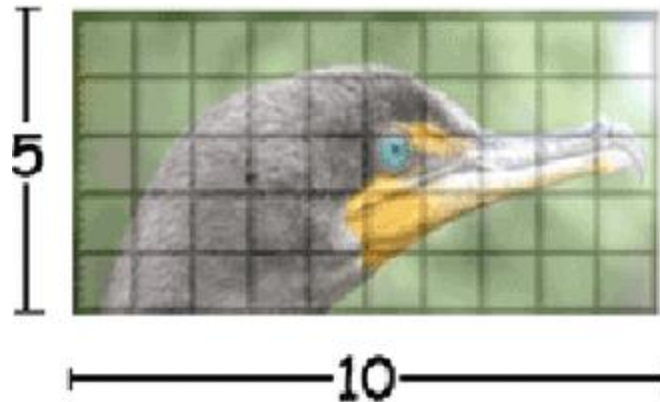
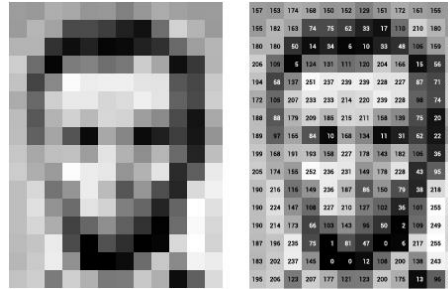


Image Representation



157	153	174	168	150	152	129	151	172	161	155	156
155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	94	6	10	33	48	105	159	181
206	109	5	124	131	111	120	204	165	15	56	180
194	58	137	251	237	239	239	228	227	87	71	201
172	105	207	233	233	214	220	239	228	98	74	205
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	105	35	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	85	150	79	38	218	241
190	224	147	108	227	210	127	102	35	101	255	224
190	214	173	66	103	143	95	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

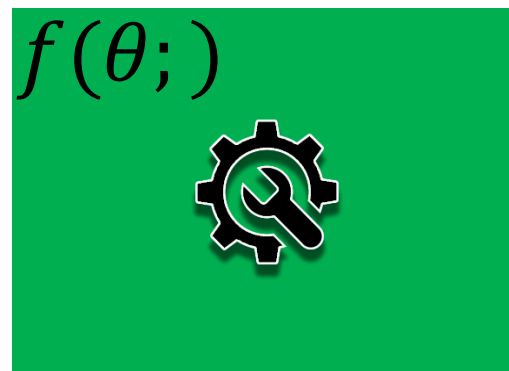
Image Representation



Example of an AI system which outputs high-level image description



Image
or
Image Representation



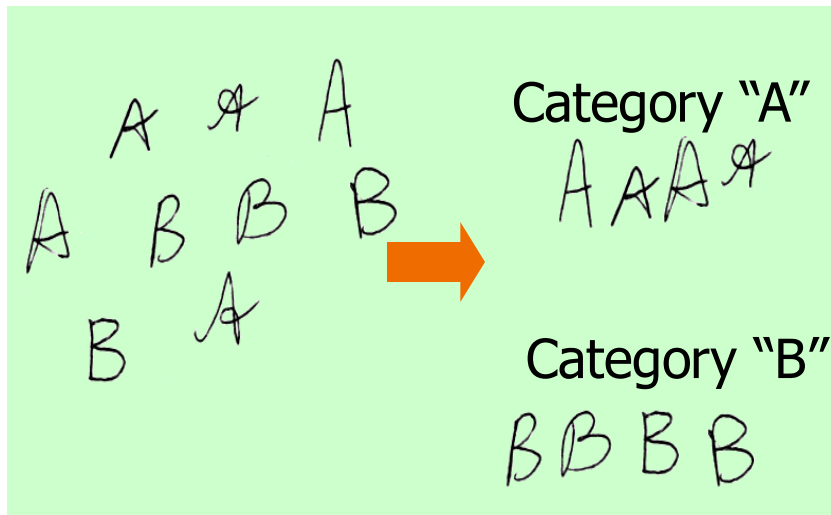
$$y = f(\theta; I)$$

Age = 32

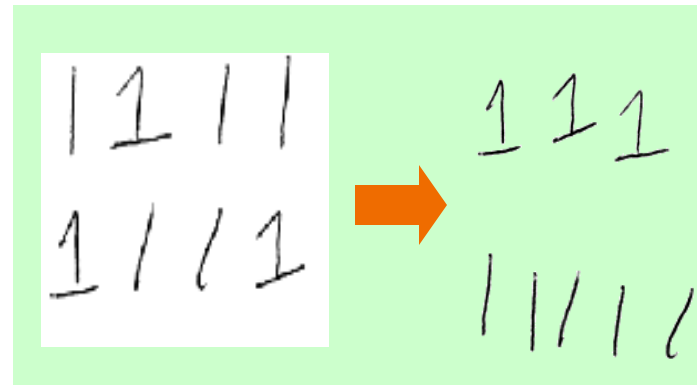
(High-level)
Image Description

Classification vs Clustering

- Classification (known categories)
- Clustering (unknown categories)



Classification (Recognition)
(Supervised Classification)



Clustering
(Unsupervised Classification)

Various kinds of ML problems involving image

Classification



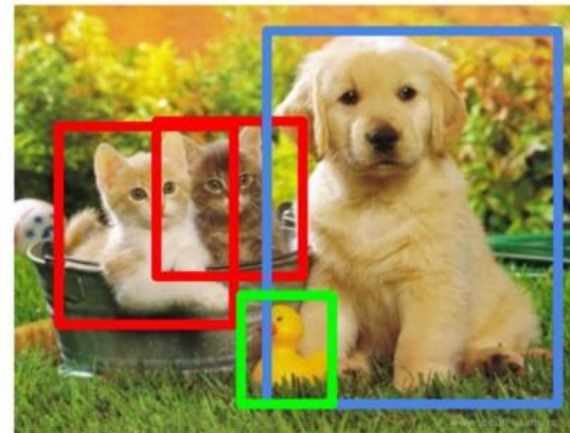
CAT

**Classification
+ Localization**



CAT

Object Detection



CAT, DOG, DUCK

**Instance
Segmentation**

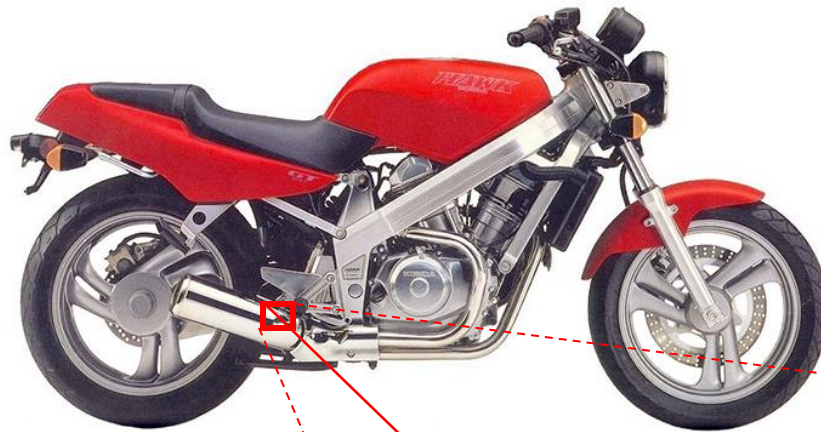


CAT, DOG, DUCK

Single object

Multiple objects

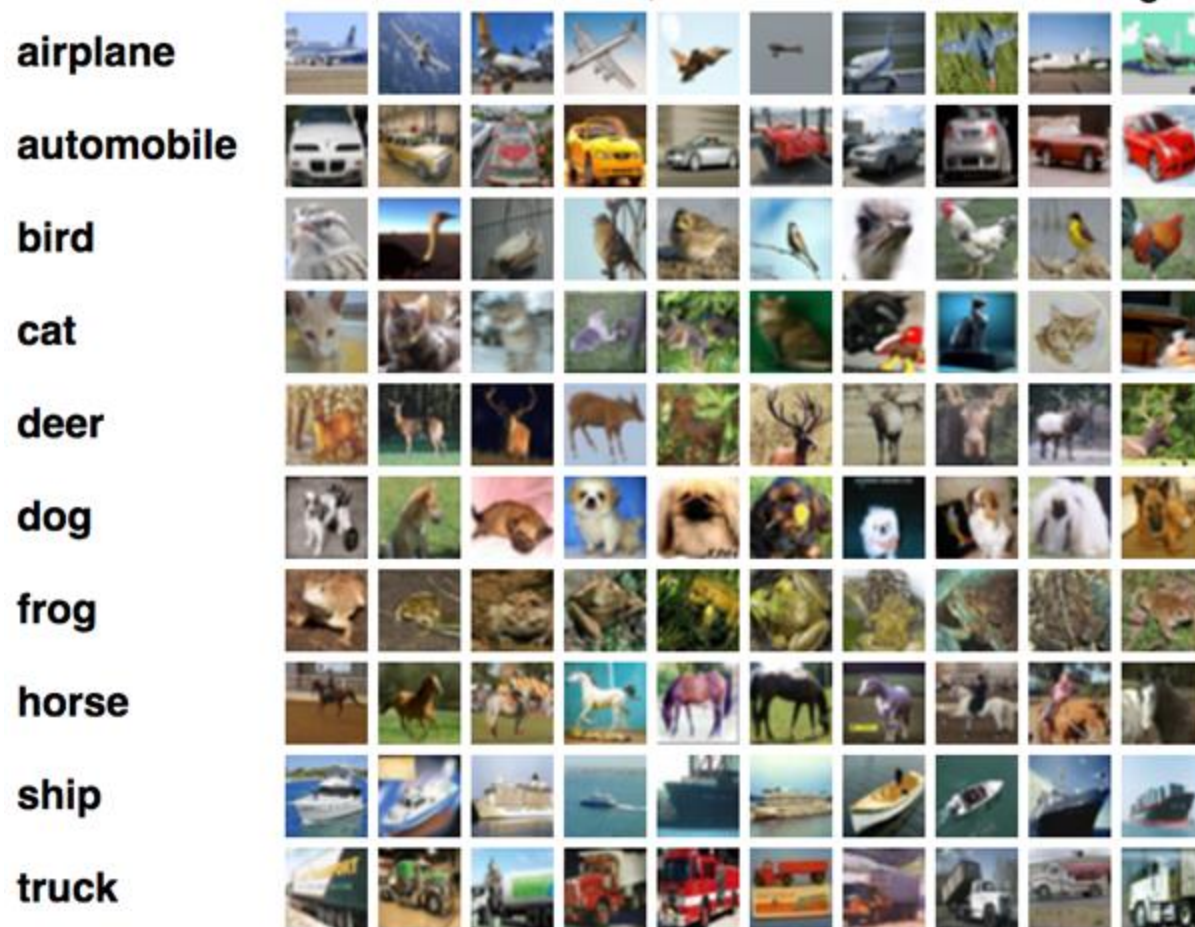
Why is this hard?



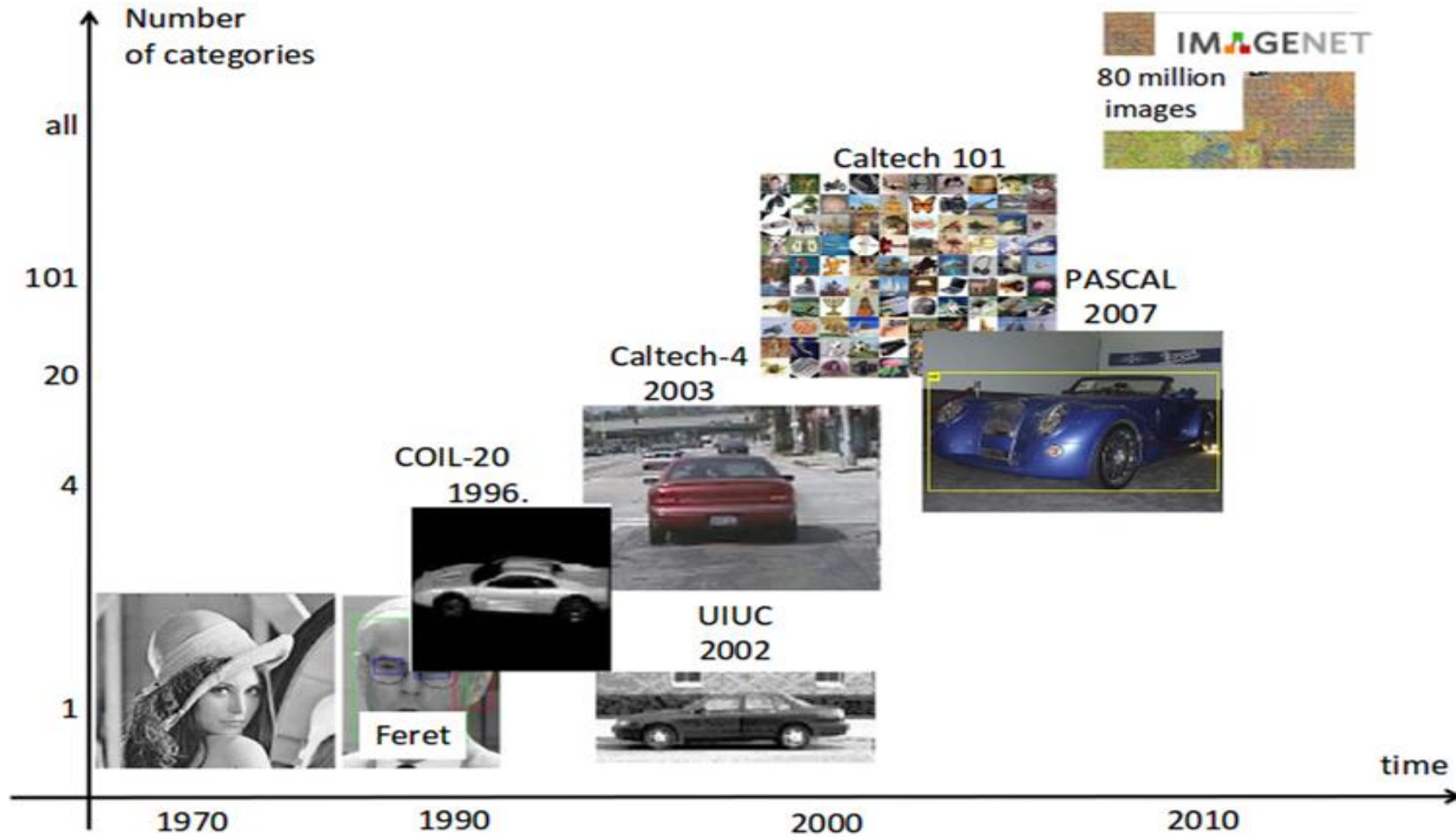
194	210	201	212	199	213	215	195	178	158	182	209
180	189	190	221	209	205	191	167	147	115	129	163
114	126	140	188	176	165	152	140	170	106	78	88
87	103	115	154	143	142	149	153	173	101	57	57
102	112	106	131	122	138	152	147	128	84	58	66
94	95	79	104	105	124	129	113	107	87	69	67
68	71	69	98	89	92	98	95	89	88	76	67
41	56	68	99	63	45	60	82	58	76	75	65
20	43	69	75	56	41	51	73	55	70	63	44
50	50	57	69	75	75	73	74	53	68	59	37
72	59	53	66	84	92	84	74	57	72	63	42
67	61	58	65	75	78	76	73	59	75	69	50

A Typical Problem: CIFAR-10

- 10 classes. 50K Train. 10K Test



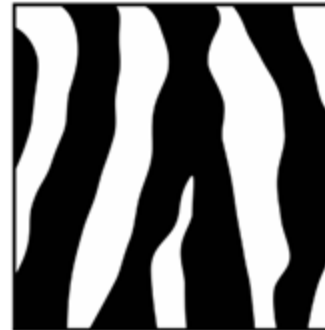
- Our “smaller” Lab problem
 - Automobile vs bird
 - Separate/recognize two classes
 - 10K Train 2K Test Samples



A Naïve Attempt

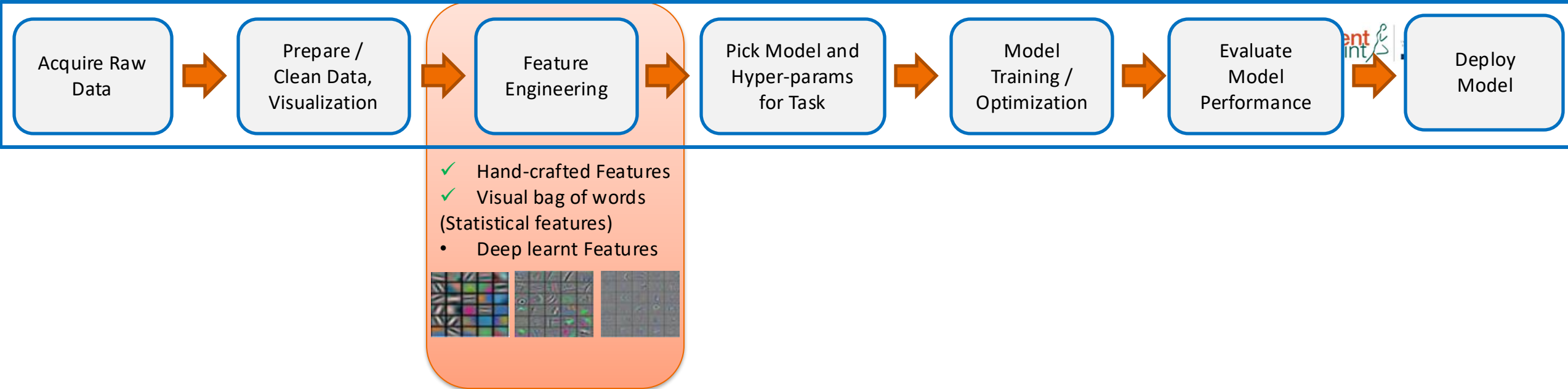
- If (image has green/grass)
 - It is an animal
 - if (it is tall) it is a horse
 - If (image has blue)
 - It is either airplane or bird
 - If (...)
 - And so on ..
- Any hope of this working? 😊

Human knowledge



Human knowledge



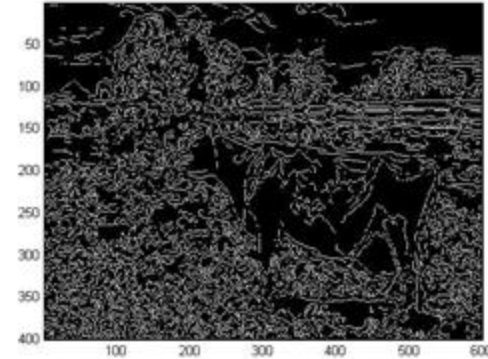


Deep Features

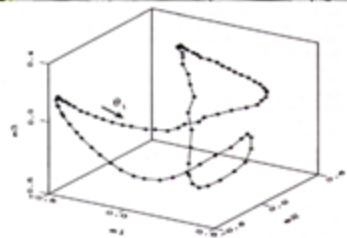
_____ Neural Network, Deep Learning and Deep _____
Features

Image Representation

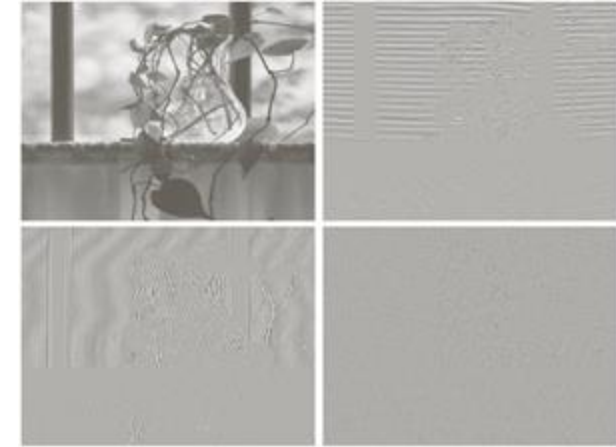
Features: Classical



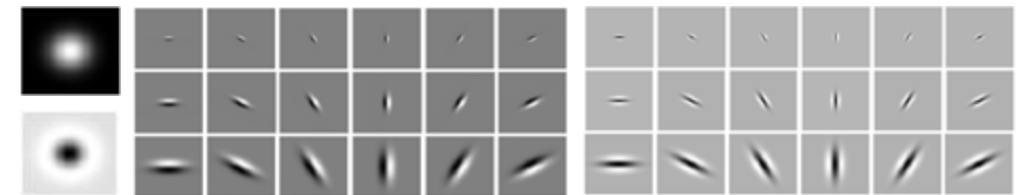
Edges and Corners: Sobel, LoG and Canny



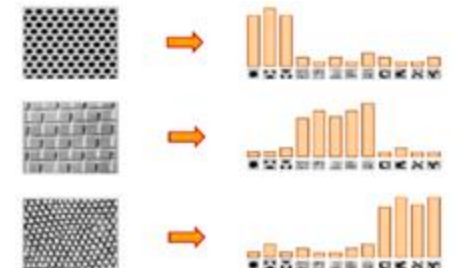
PCA, Subspaces and Manifolds



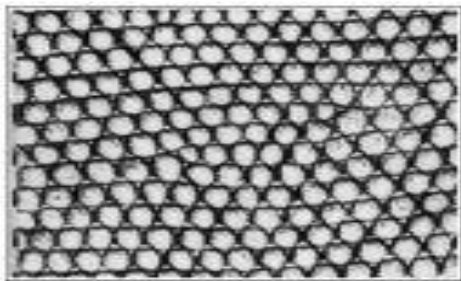
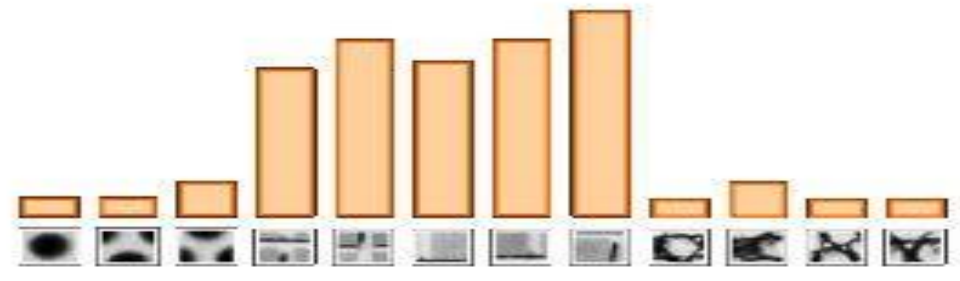
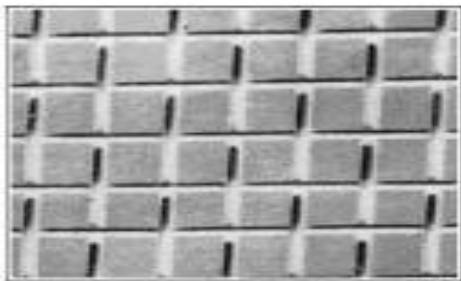
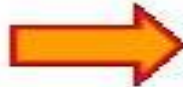
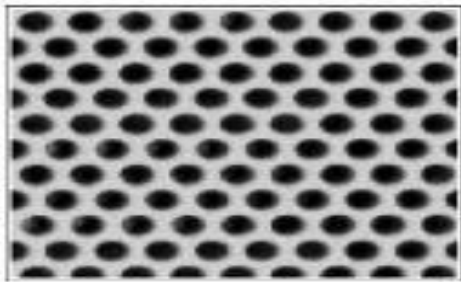
Fourier and Wavelet



Texture; Filter bank; Histogram of responses

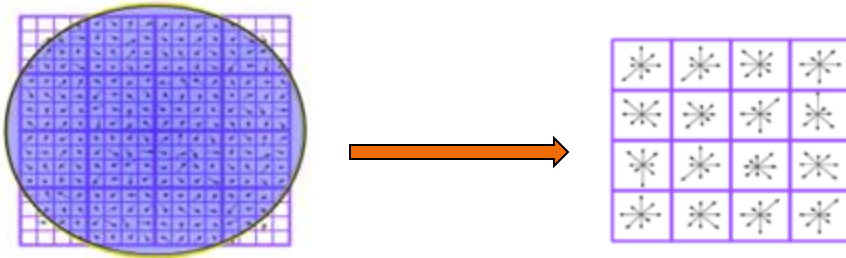


Visual BoW: Histogram of Parts

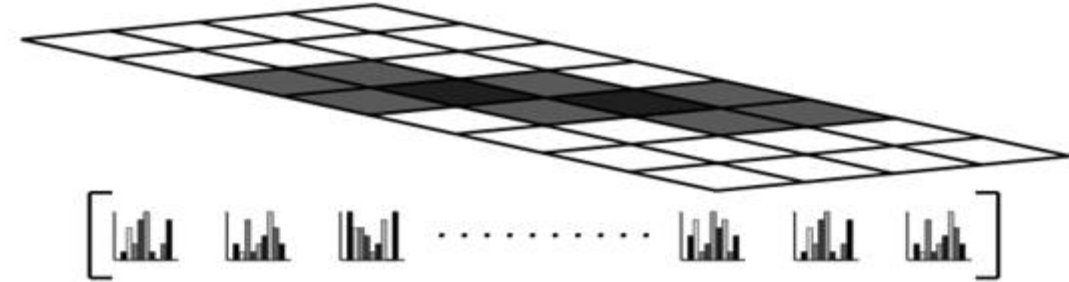


Source: wordpress.com/2013/04/24/research-bag-of-features-for-visual-recognition

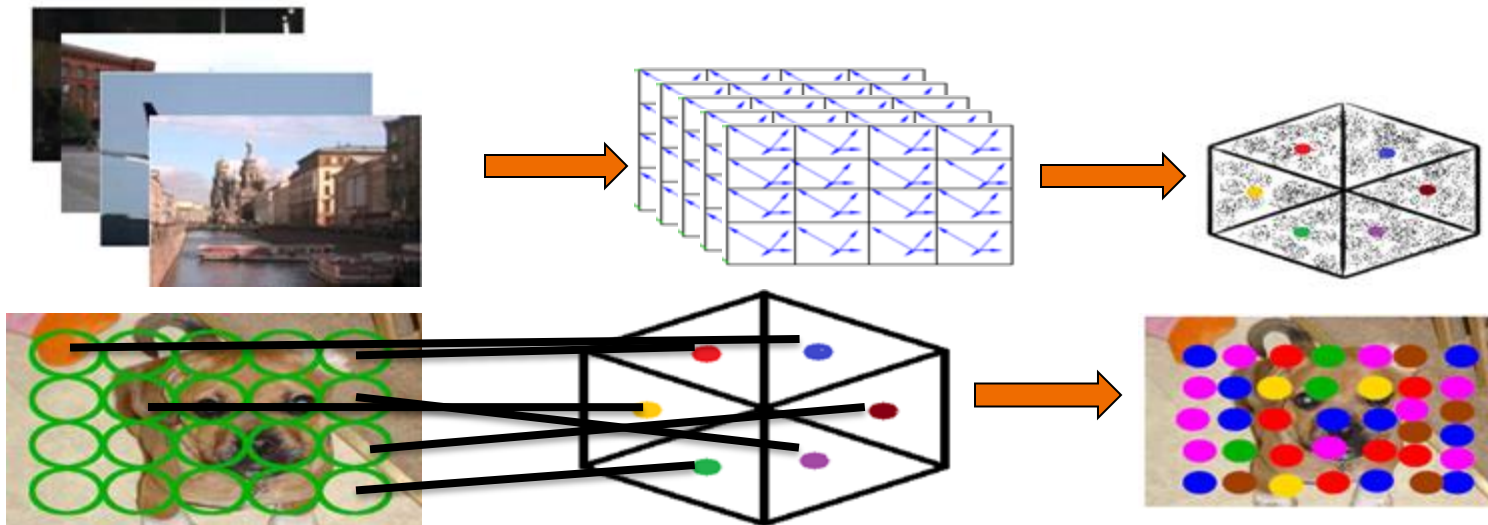
Well Engineered Features



SIFT (Lowe 1999, 2004)



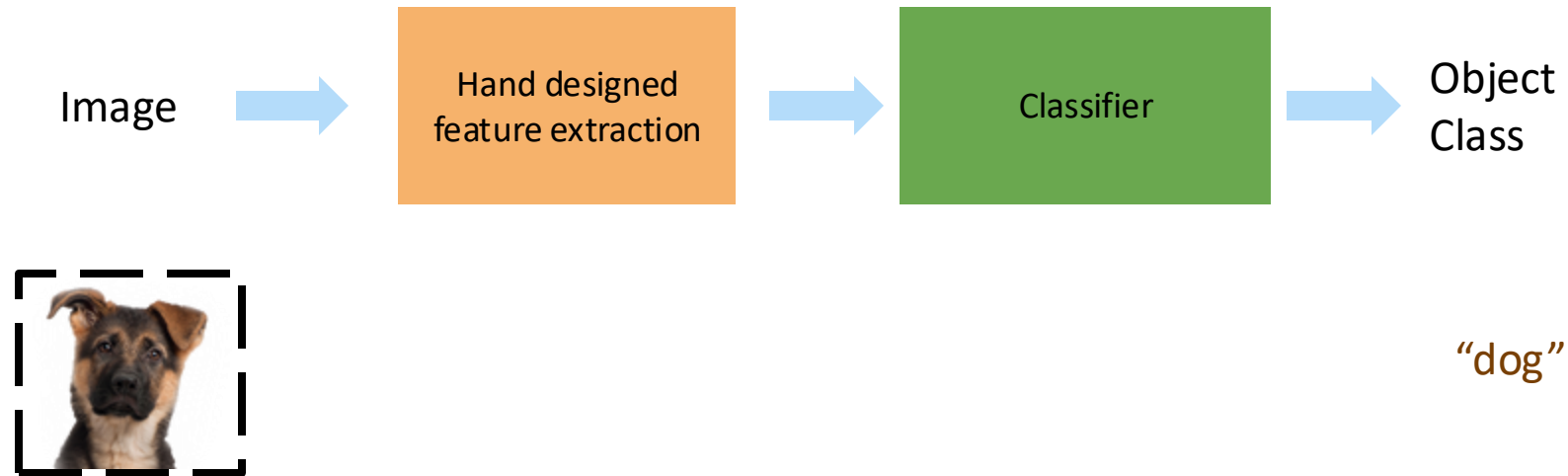
HOG (Dalal and Triggs 2005)



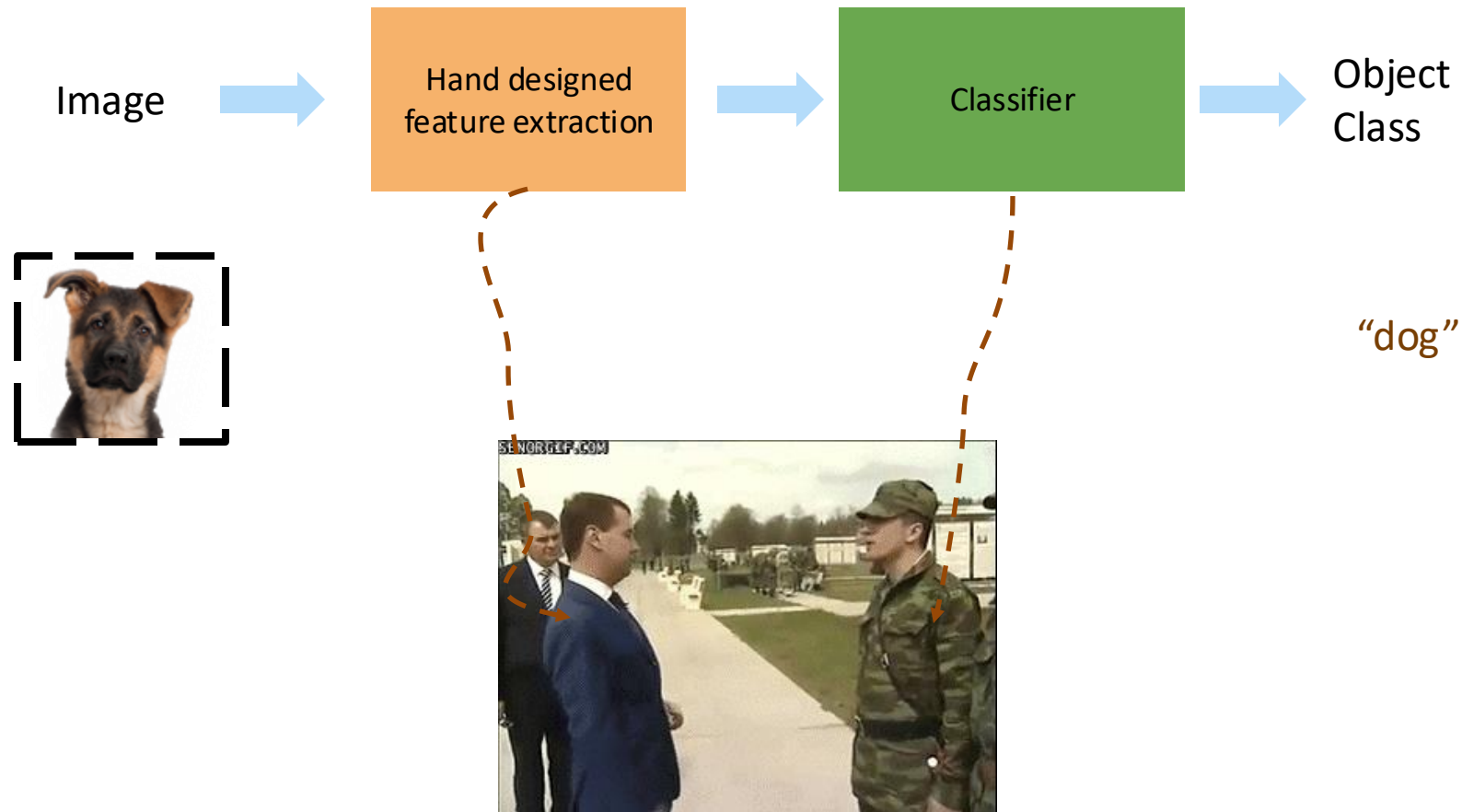
Bag of Words (Sivic and Zisserman 2003)

*Focus: Dictionary Learning,
Pooling and Coding*

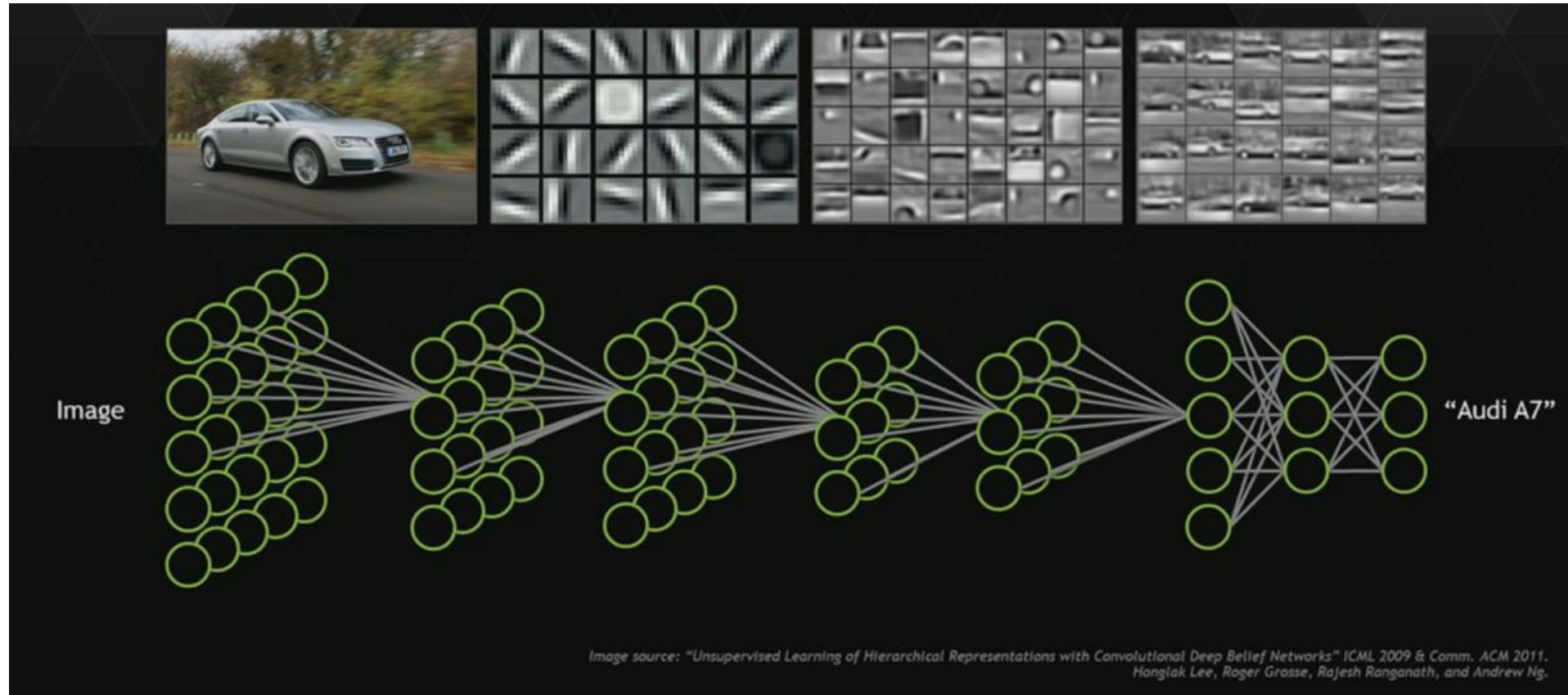
Object-recognition: conventional approach



Object-recognition: conventional approach

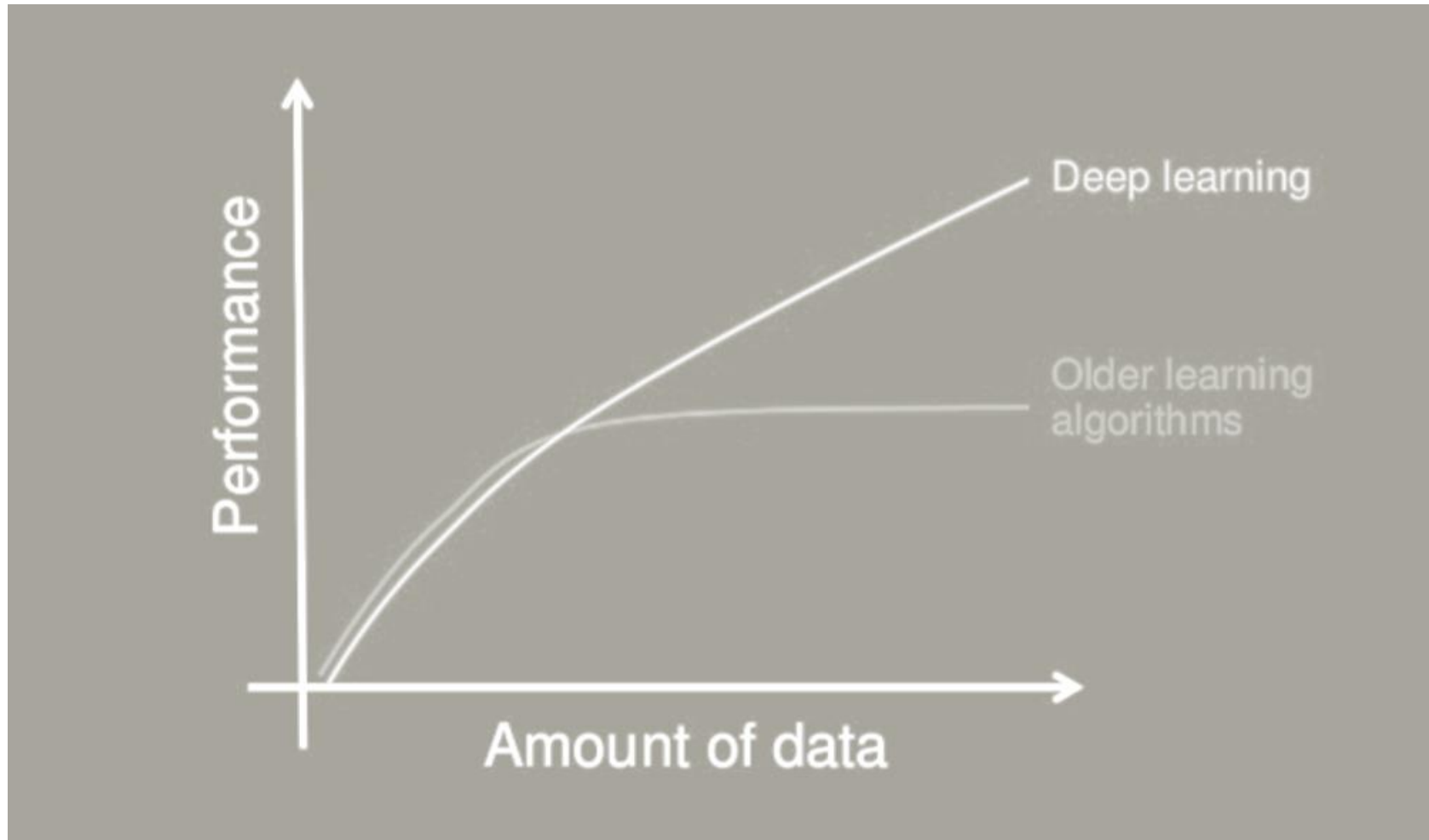


Object Recognition: Deep Neural Networks

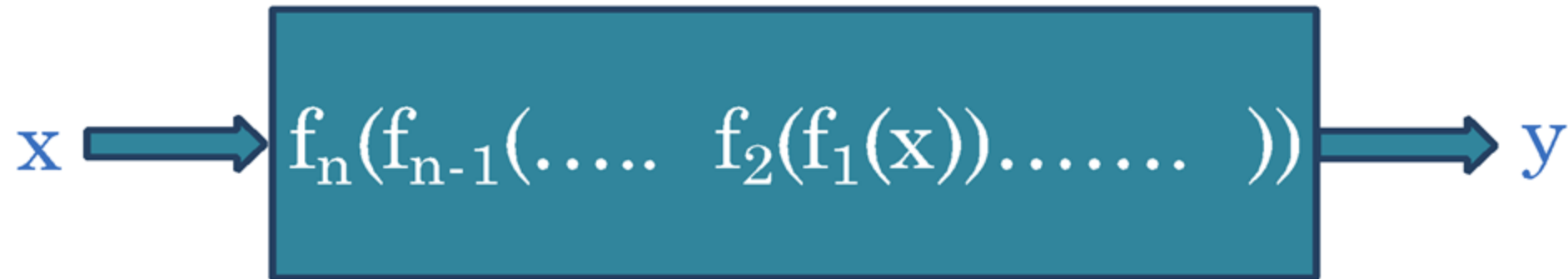


Data-driven, End-to-End learning, Task-specific feature hierarchy

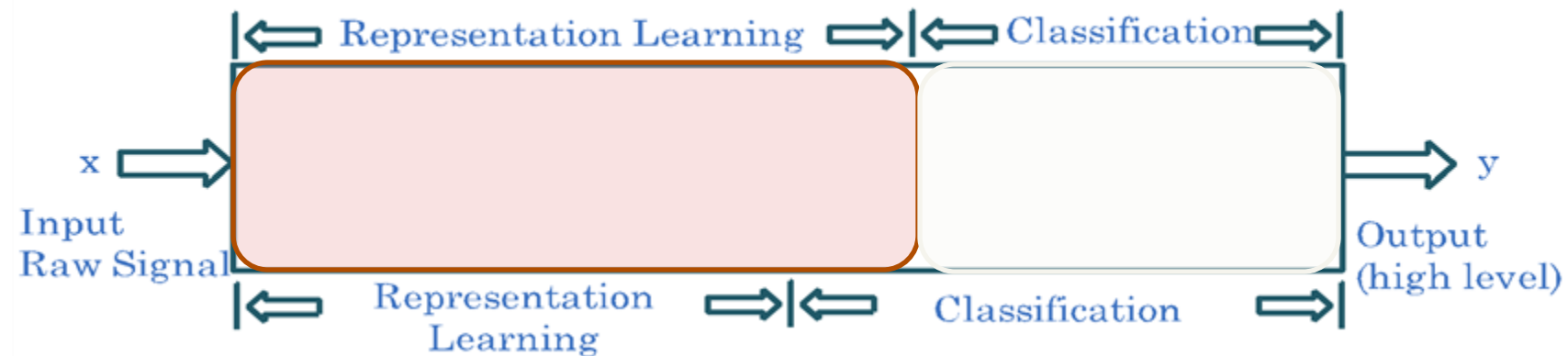
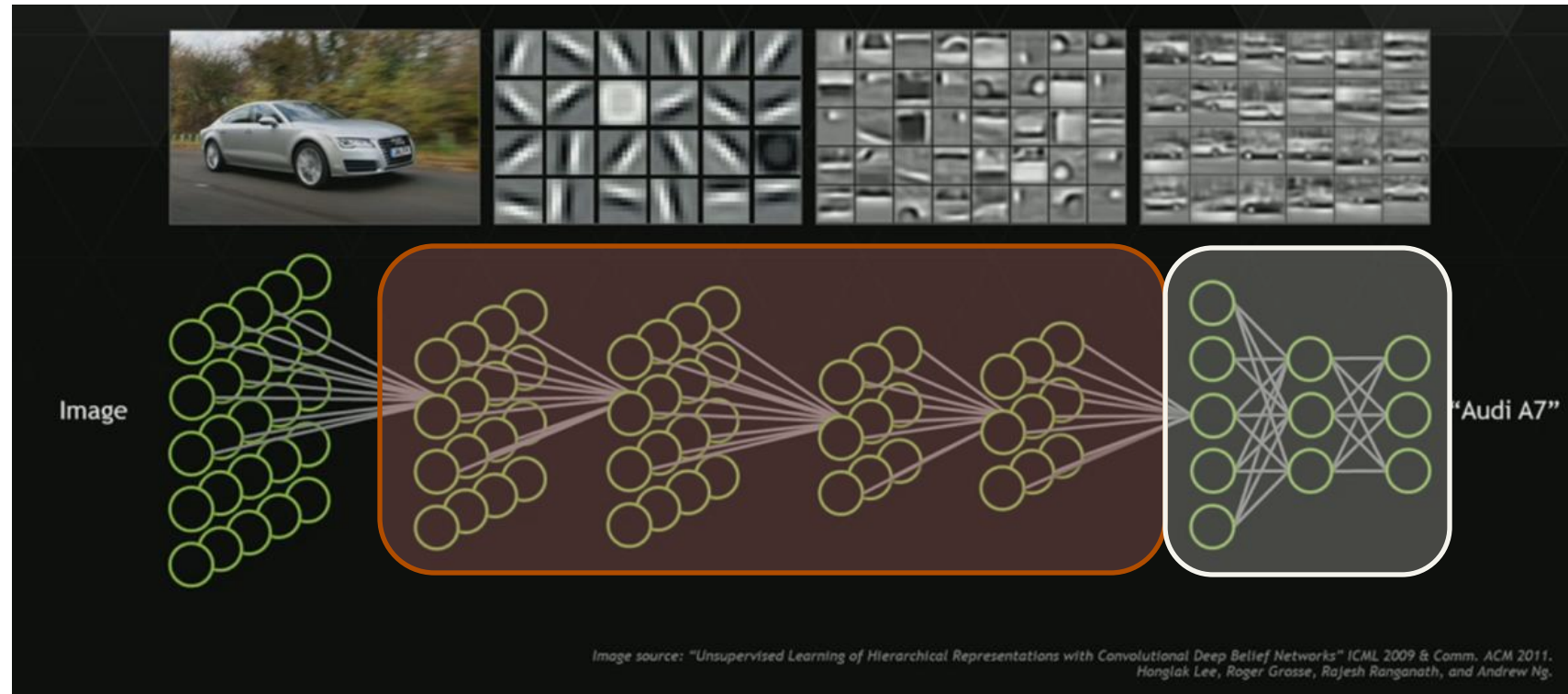
Why deep learning



Complex Functions and Richer Features

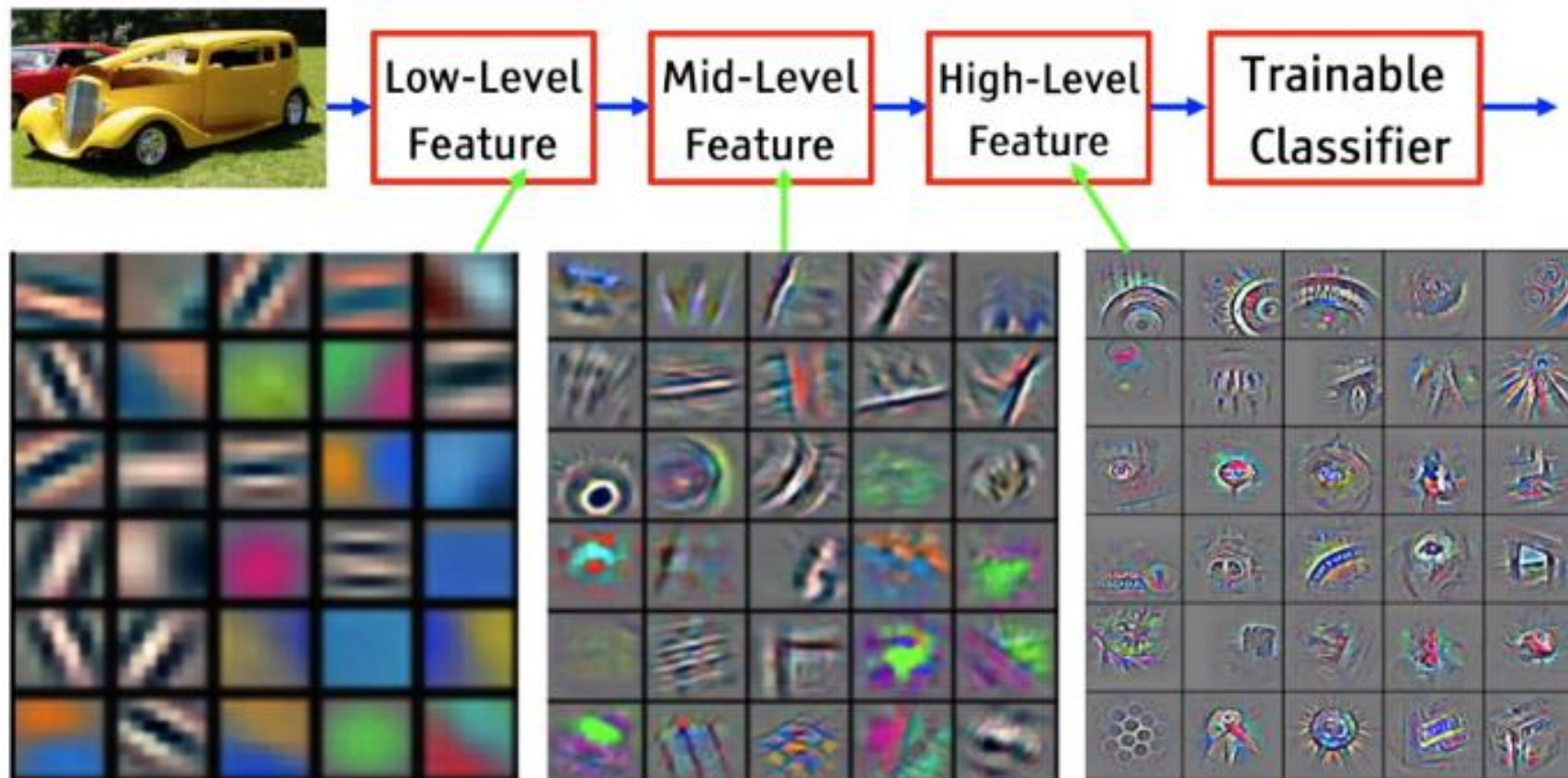


Summary



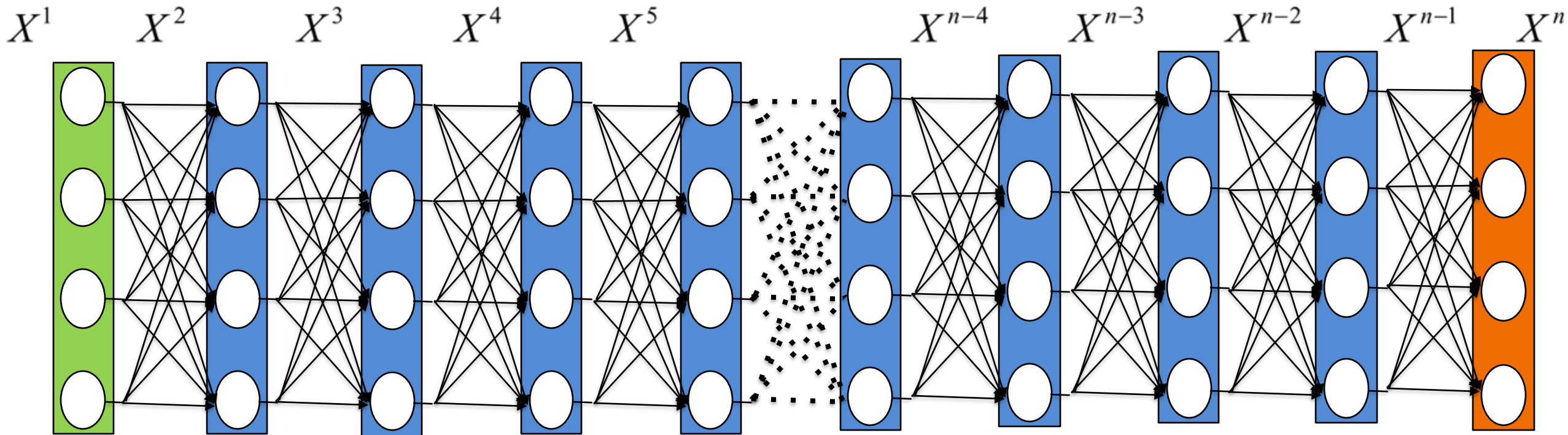
Deep Learnt Features (2013-XXX)

- It's **deep** if it has **more than one stage** of non-linear feature transformation.



Deep Features

Cutting a Trained Neural Network



Input

Hidden
layers

Output

Cut after 2nd Hidden Layer

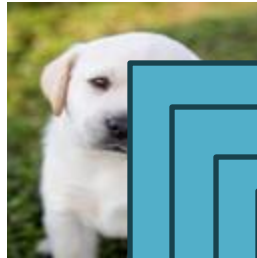
Cut after (n-2)th Hidden Layer

MLP Perspective

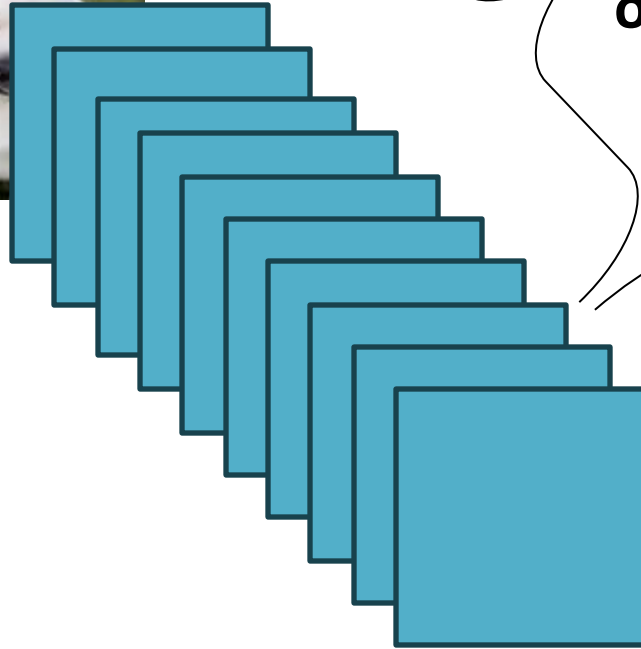
Deep Image Features

- Deep Learning = End to End Learning (Raw data to labels)
- Deep Learning = Feature Learning!!

I/P: Raw
Images



An intermediate representation from a popular “Deep VGGNet”, which was designed and trained for solving a “general” 1000 class classification.

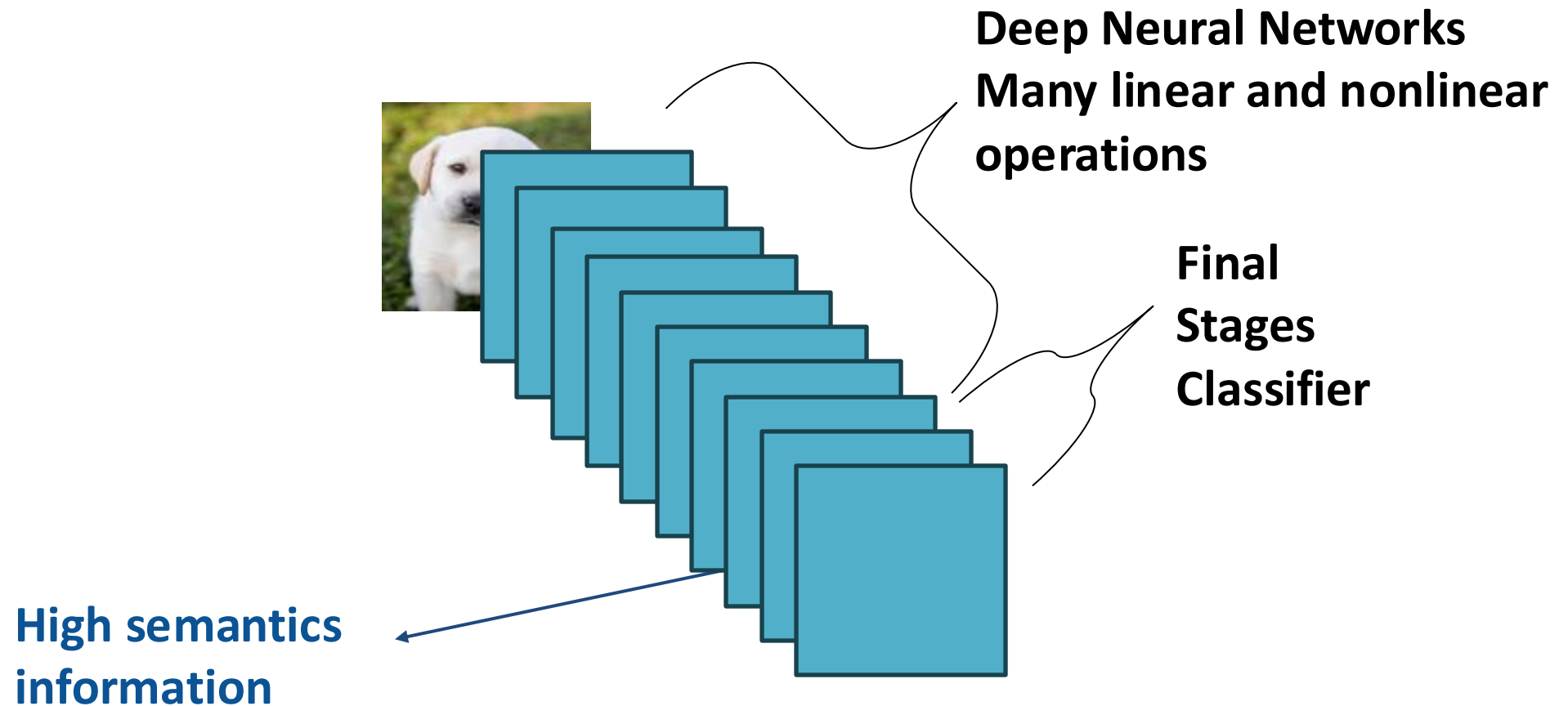


Deep Neural Networks
Many linear and nonlinear
operations

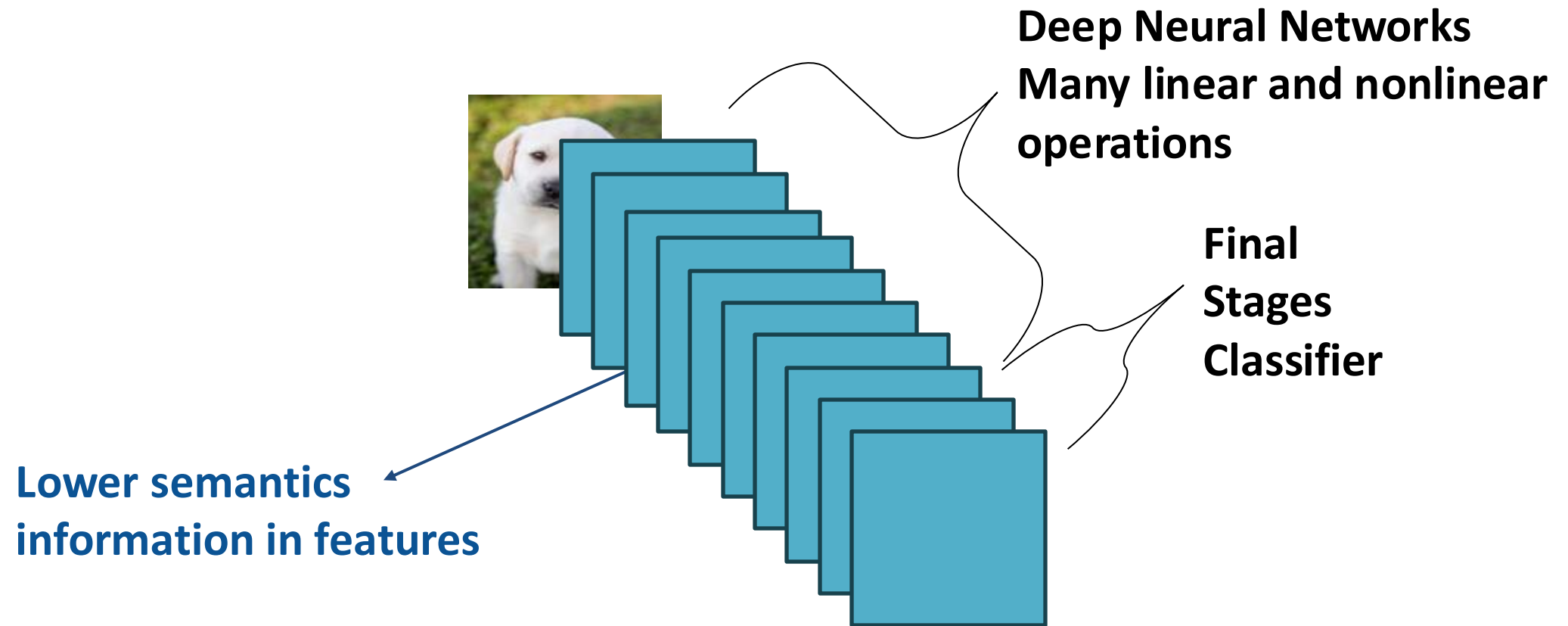
Final
Stages
Classifier



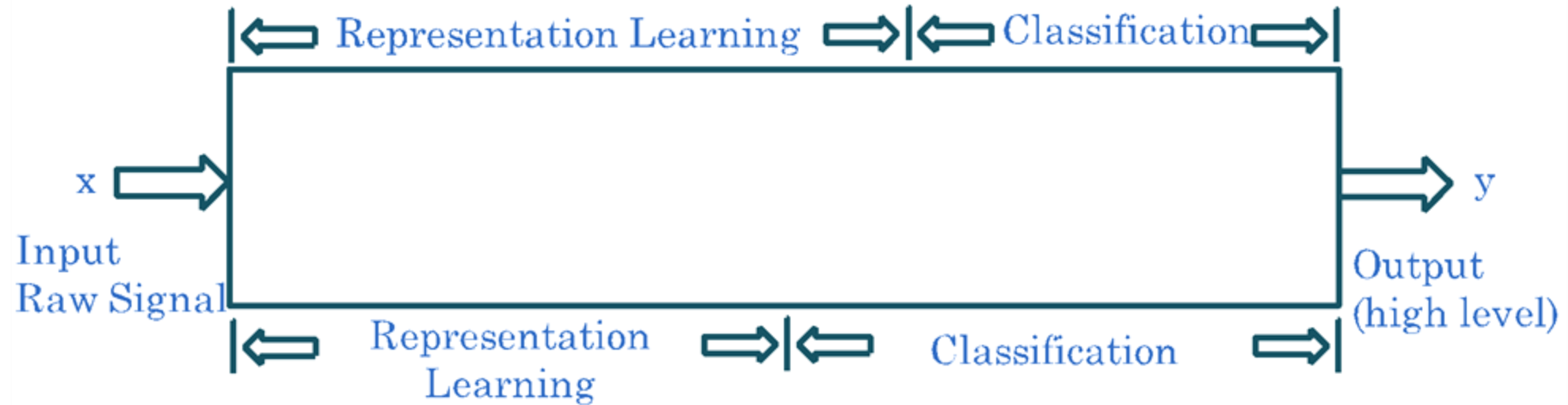
Deep Image Features

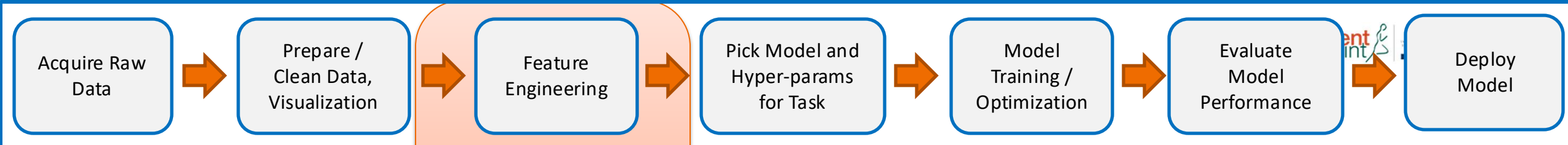


Deep Image Features



Summary





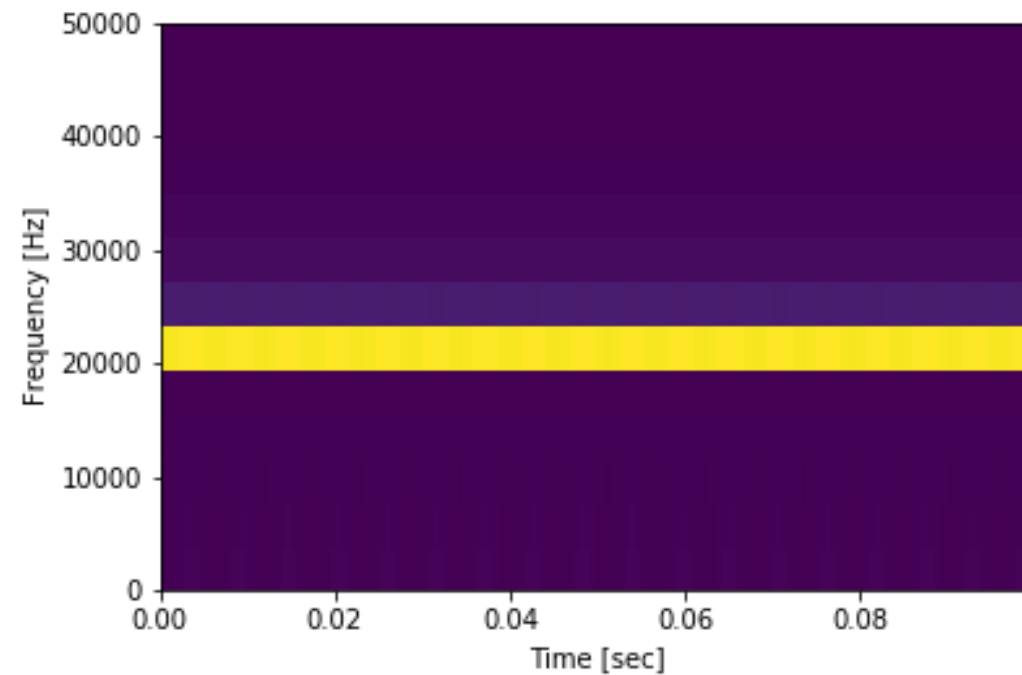
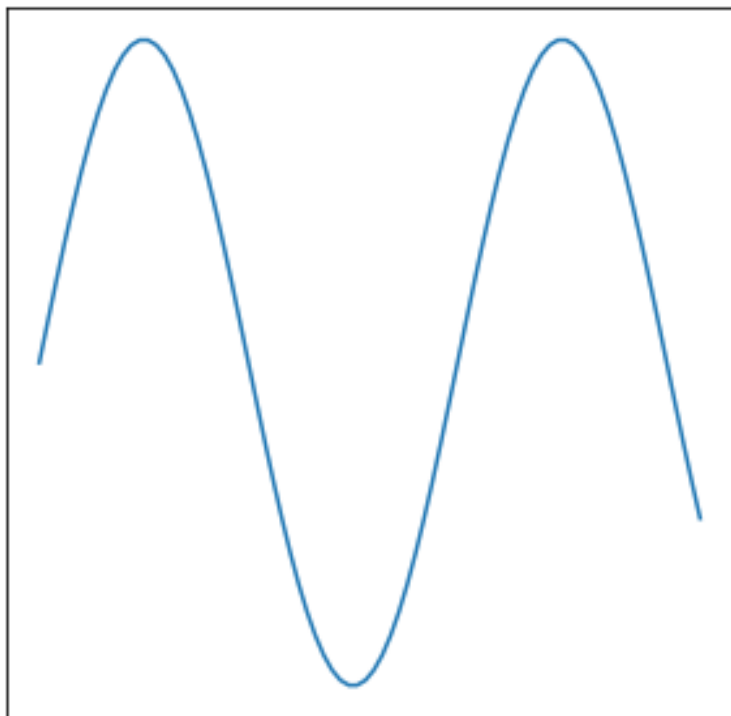
- Speech



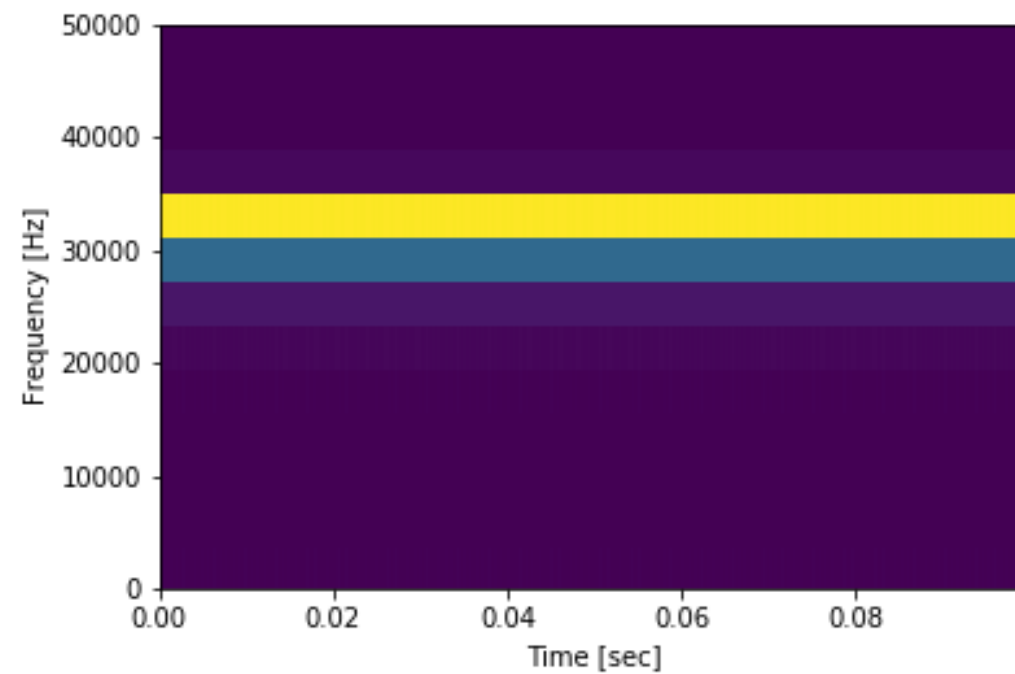
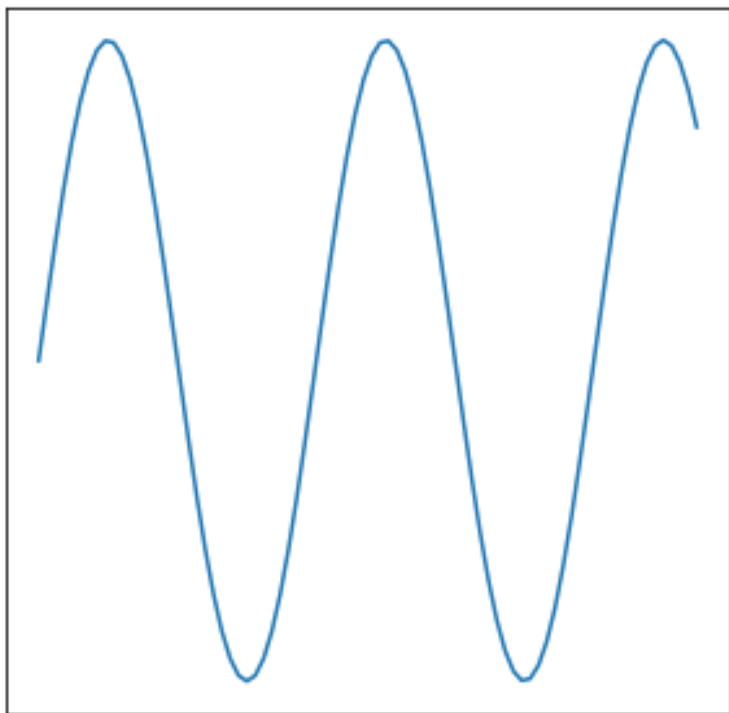
Speech

(Brief Explanation)

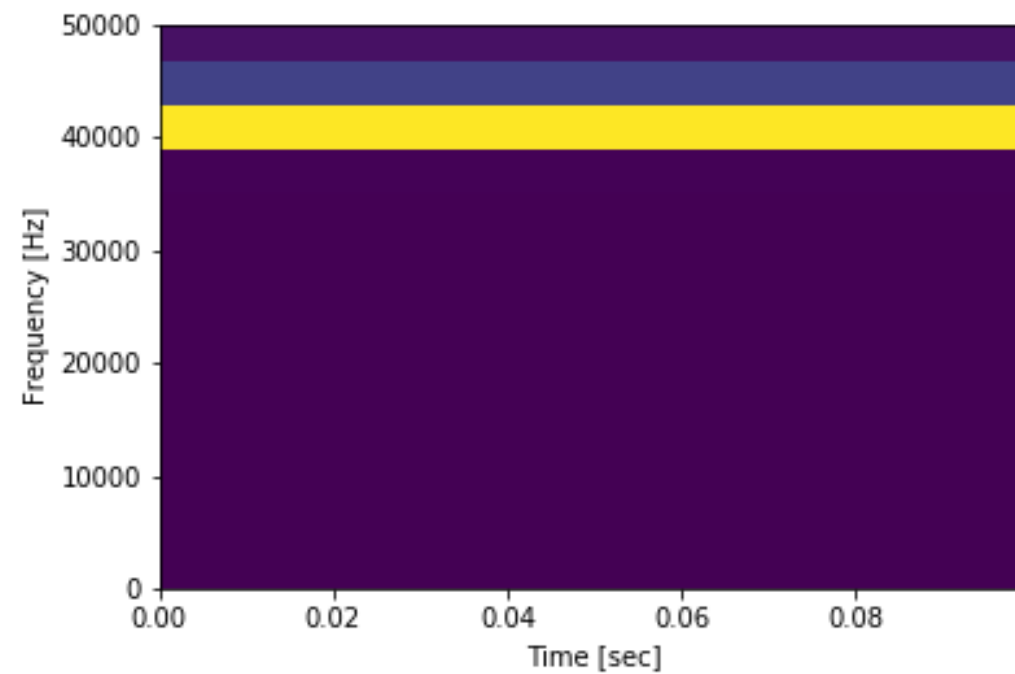
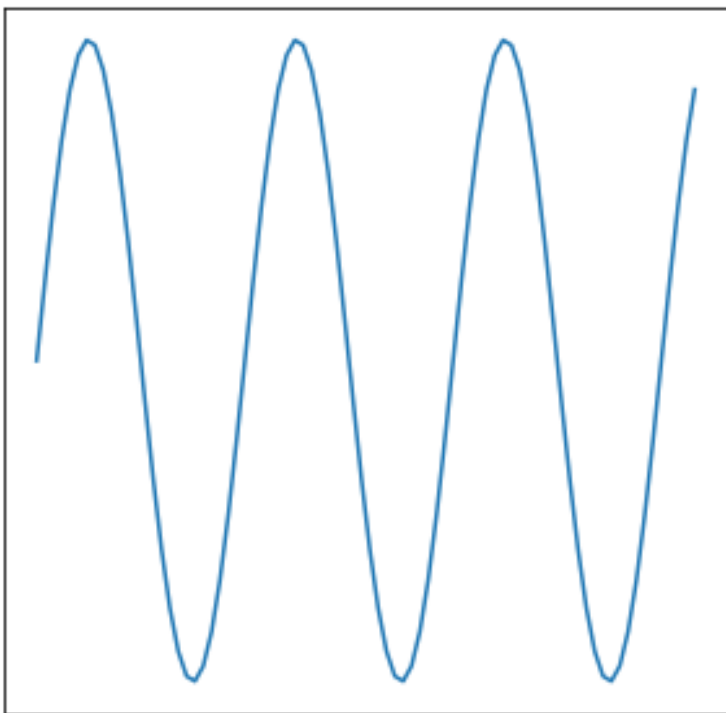
Sine wave ($f = 20\text{KHz}$)



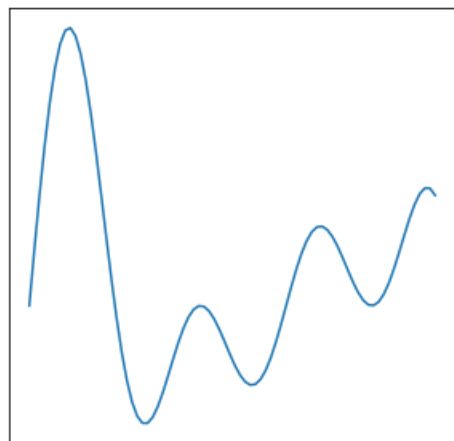
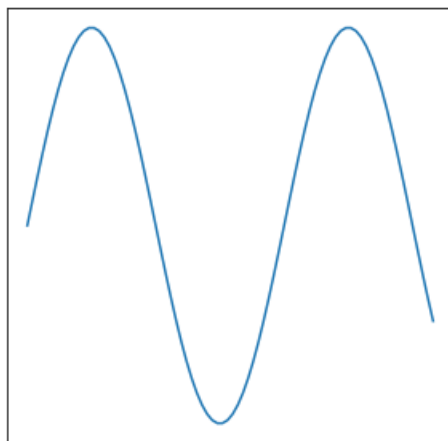
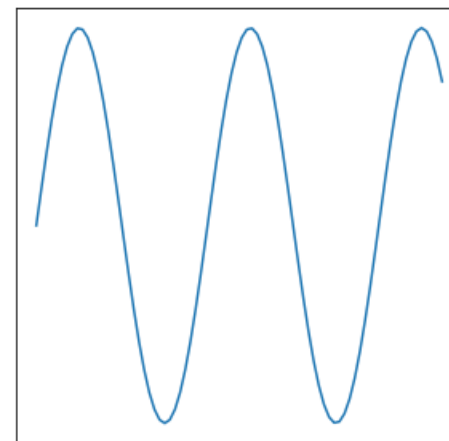
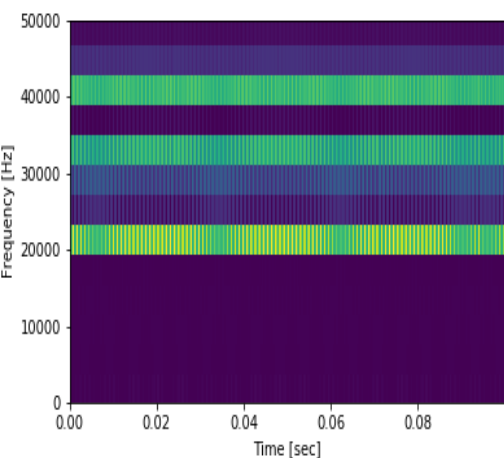
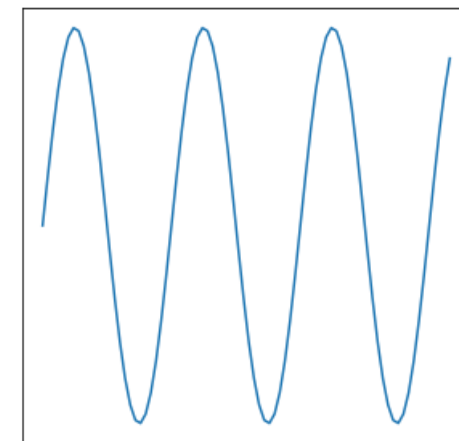
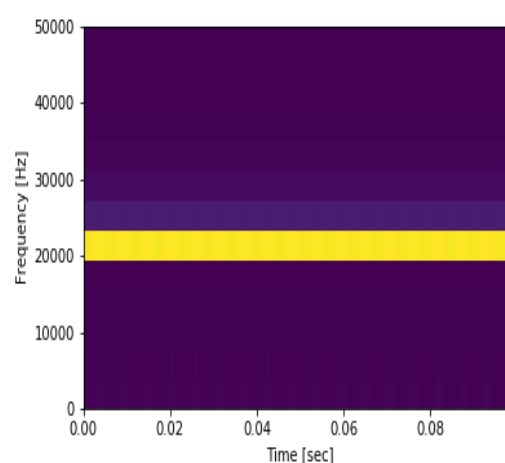
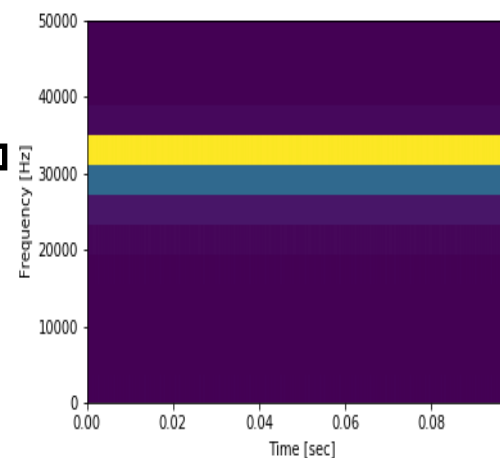
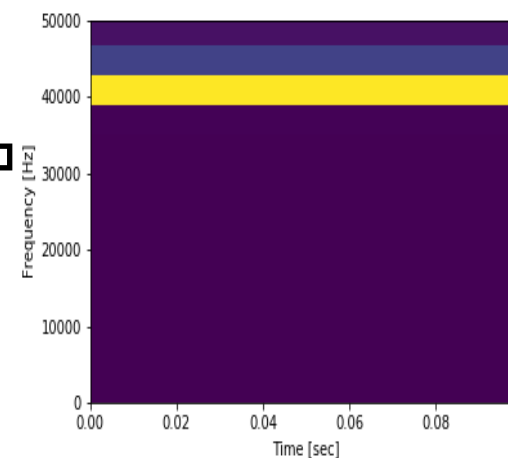
Sine wave ($f = 30\text{KHz}$)



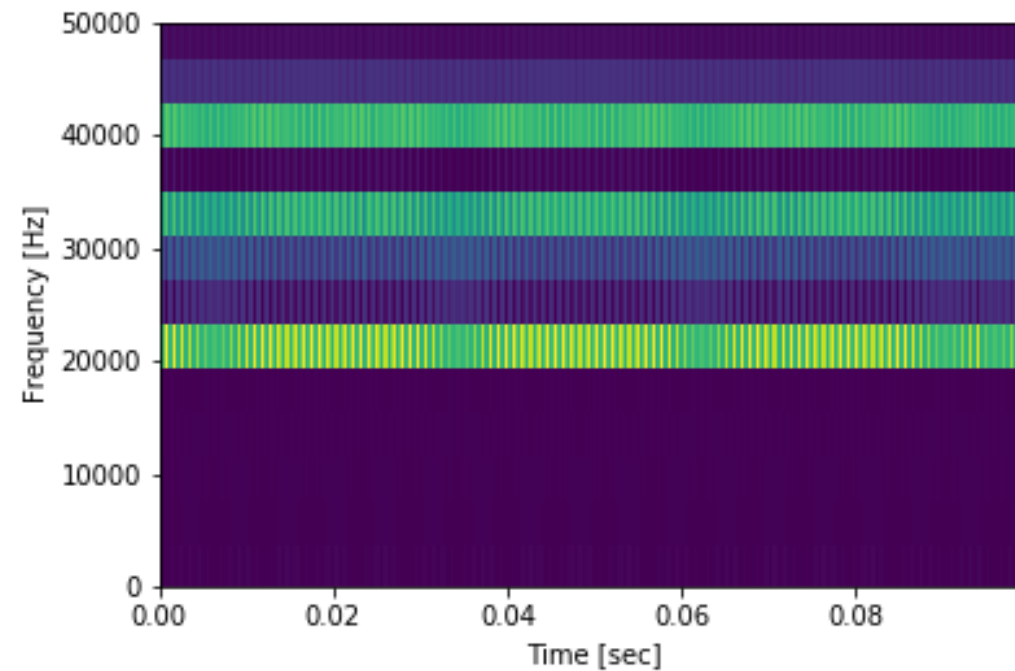
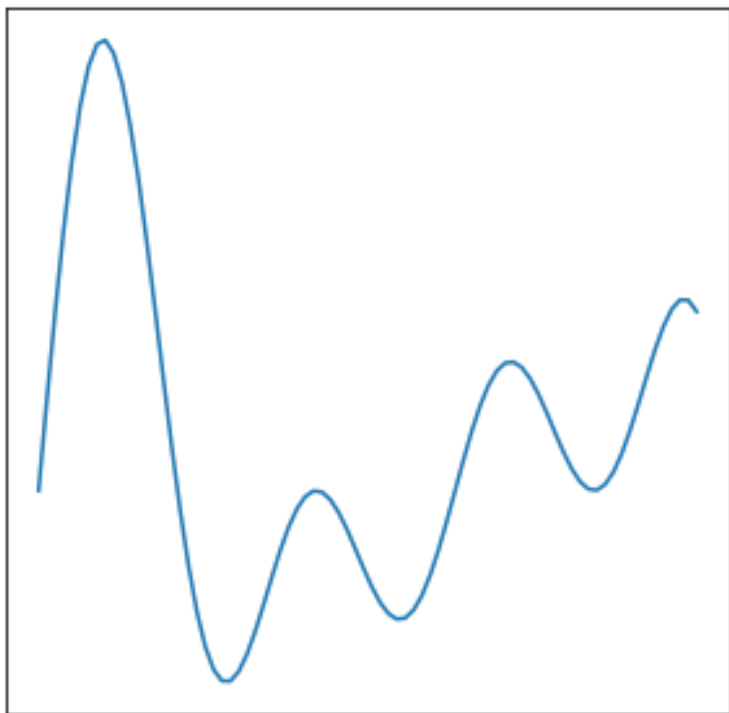
Sine wave ($f = 40\text{KHz}$)



Any wave is a combination of many sine waves

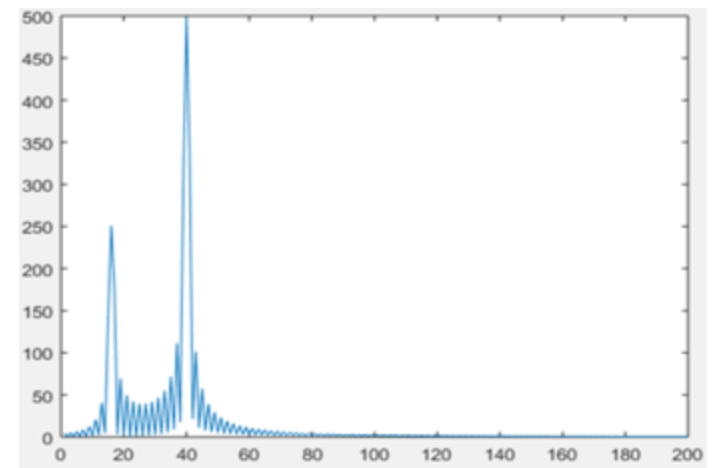
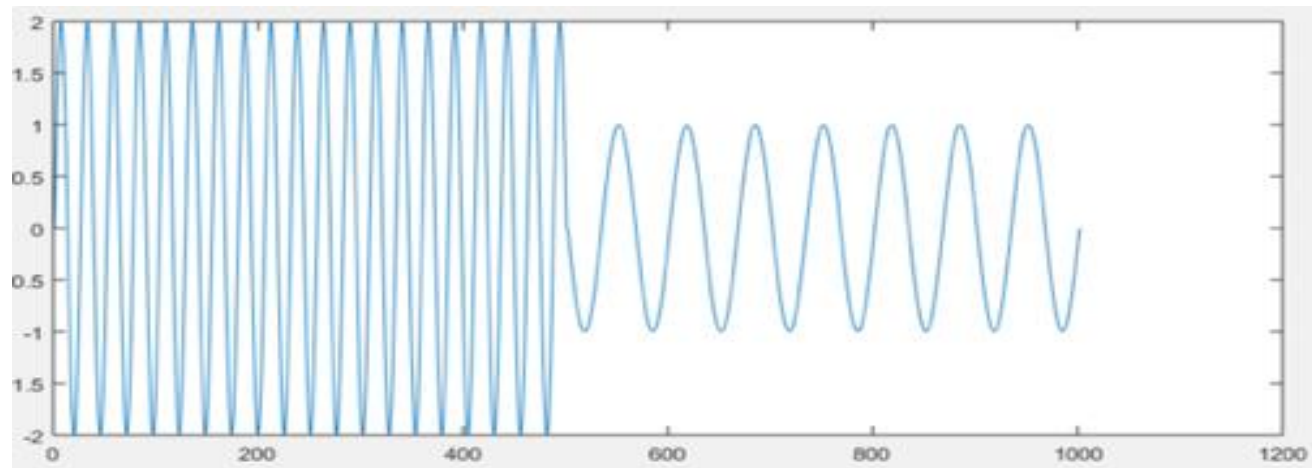
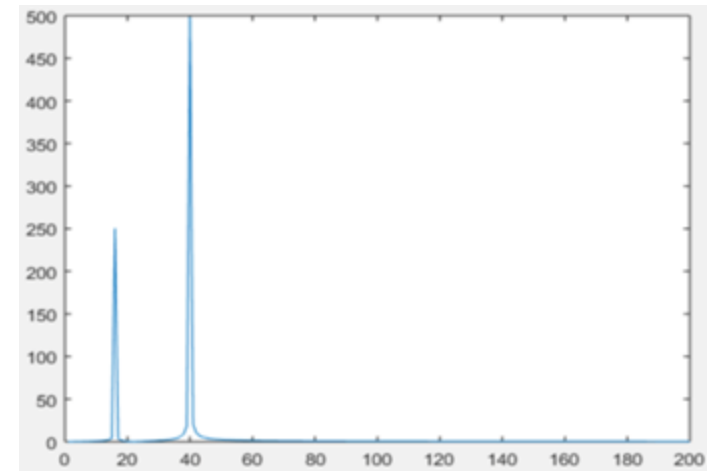
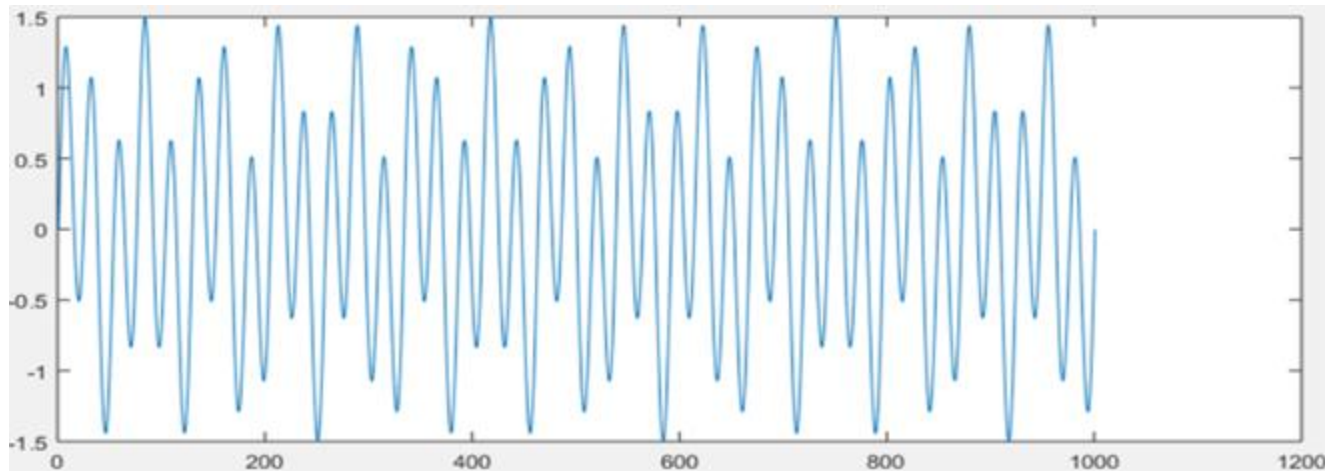
 $=$  $+$  $+$  $=$  $+$  $+$ 

Any wave is a combination of many sine waves



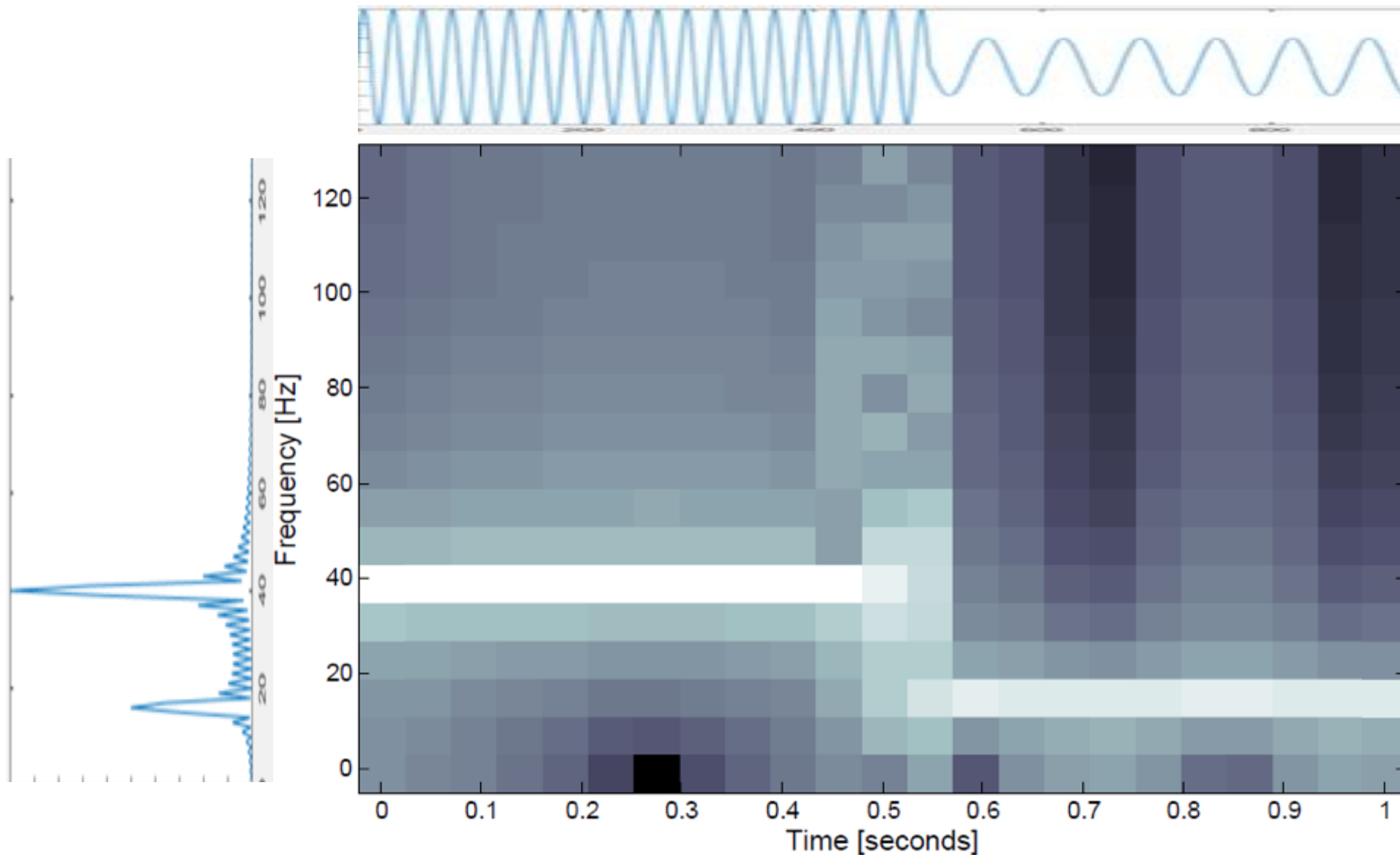
Lets understand Sound Signal

$$f(t) = \sin(2\pi \cdot 39t) + 0.5 \sin(2\pi \cdot 15t)$$



$$g(t) = \begin{cases} 2 * \sin(2\pi \cdot 39t), & 0 \leq t \leq 1/2 \\ \sin(2\pi \cdot 15t), & 1/2 < t \leq 1 \end{cases}$$

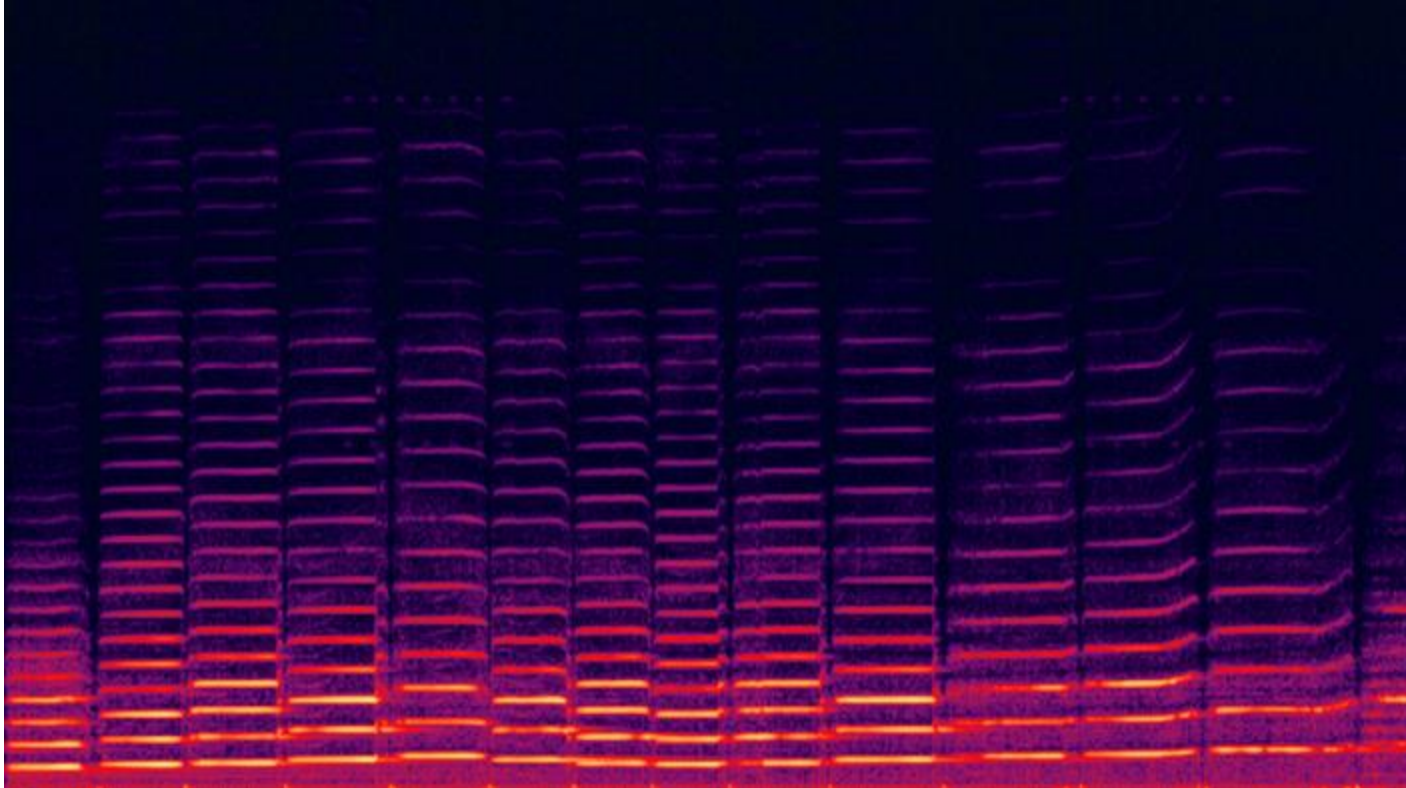
Spectrogram: Histograms of Speech Windows



Spectrogram of a
piecewise
monochromatic
signal.

Lighter color =>
greater DFT
magnitude

Spectrogram: A Simple Sound Sequence



A quick primer on sound

- <https://www.youtube.com/watch?v=jveKIYyafaQ>

Representations

A: MFCC (Signal processing based; Classical)

- Mel Frequency Cepstral Coefficients

B: CNN Based (Modern)

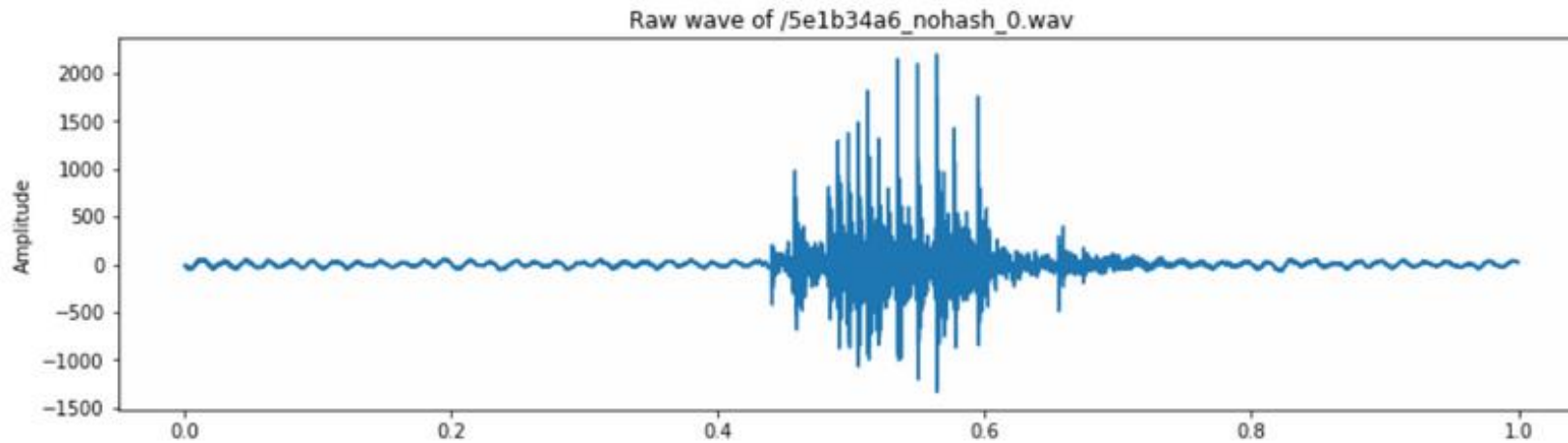
- VGG Features on the Mel Spectrogram

<http://practicalcryptography.com/miscellaneous/machine-learning/guide-mel-frequency-cepstral-coefficients-mfccs/>

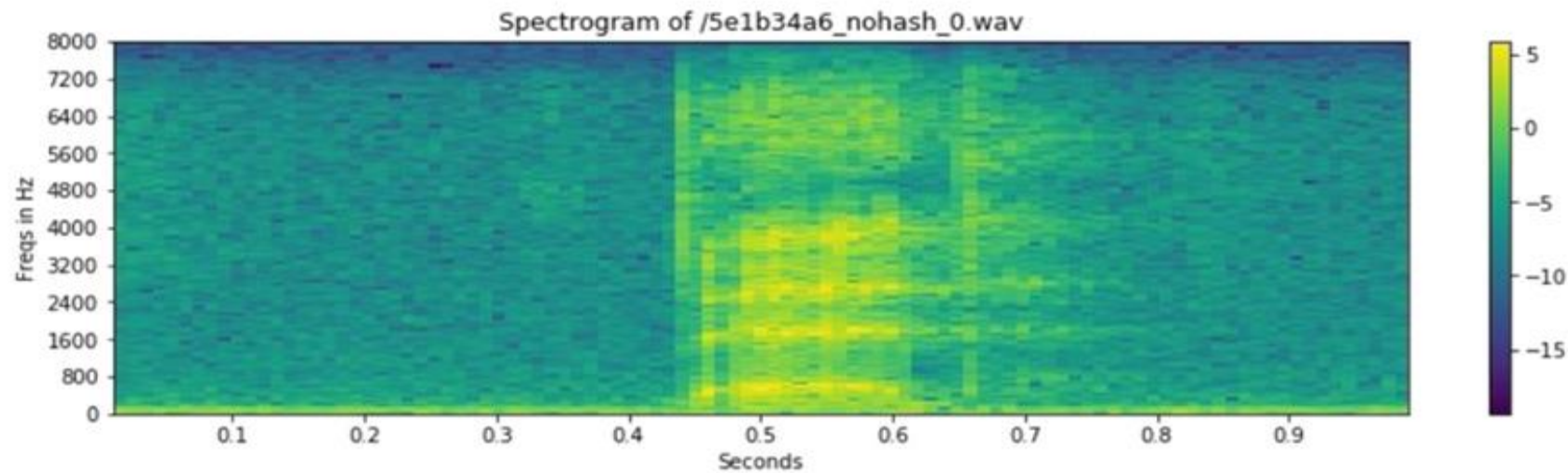
Example Problem

- Sound waves (.wav files)
- 10 short commands (“zero”, “one”, “two”)
- 1 sec duration
- 5000 samples (many people)

Classical Feature (MFCC)

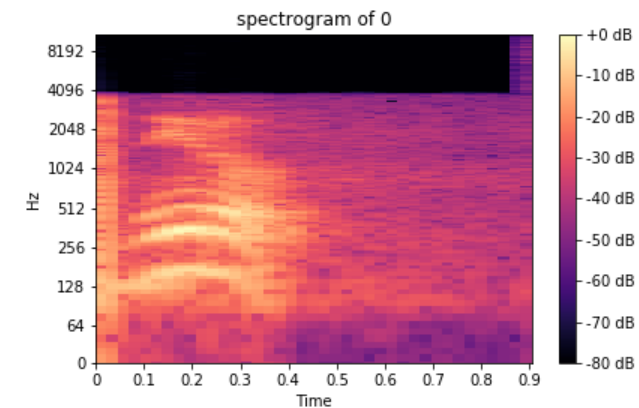
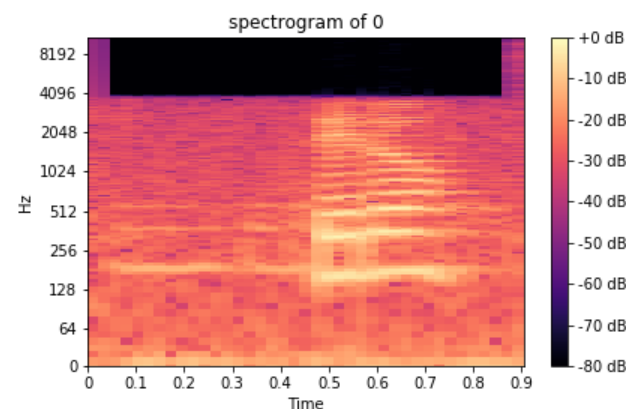
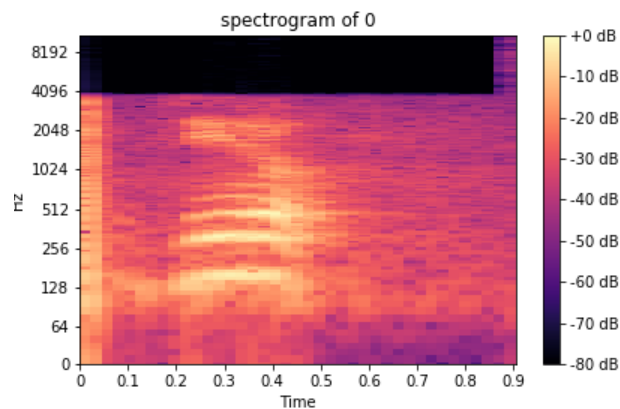
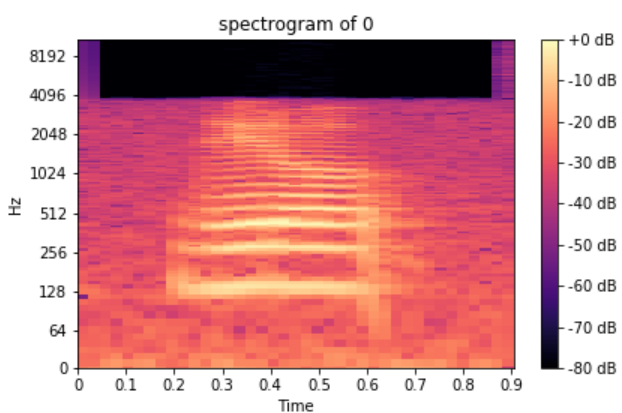


Amplitude
Vs
Time

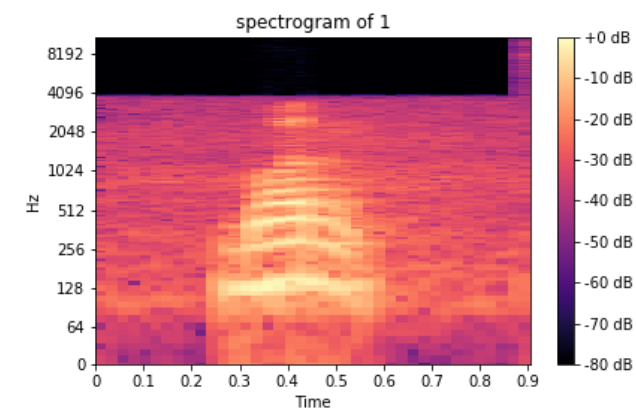
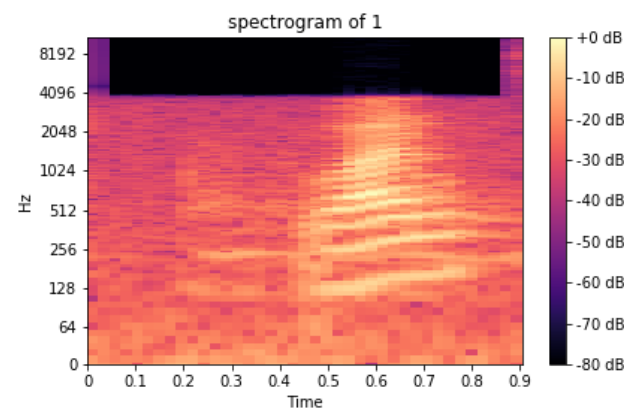
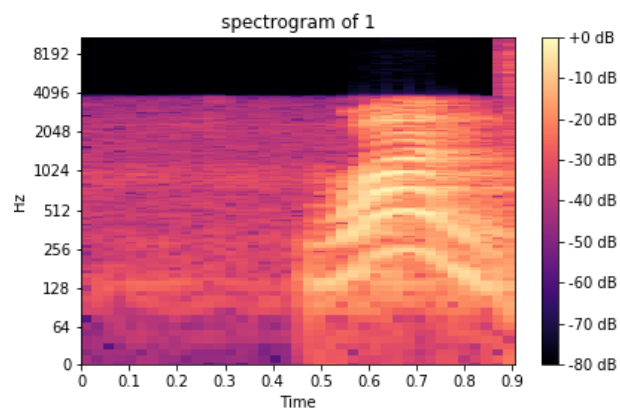
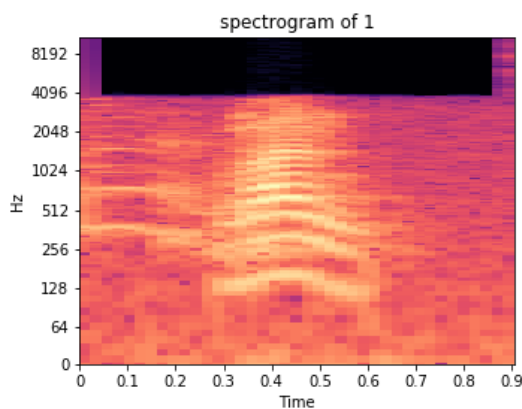


Frequency
Vs
Time

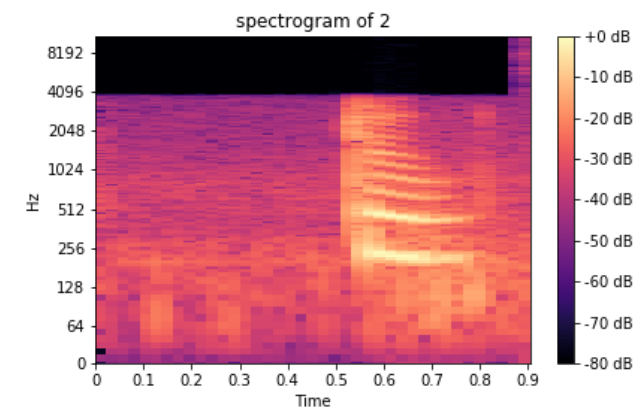
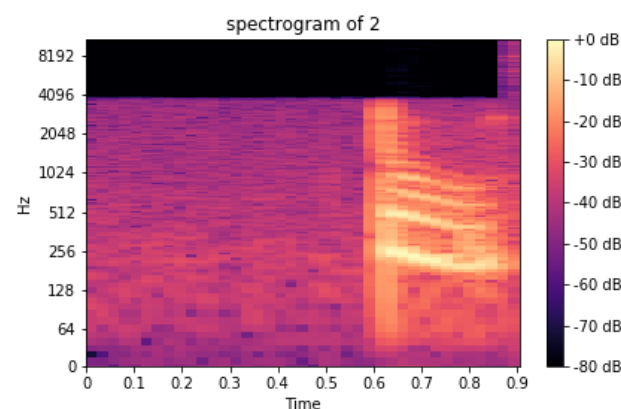
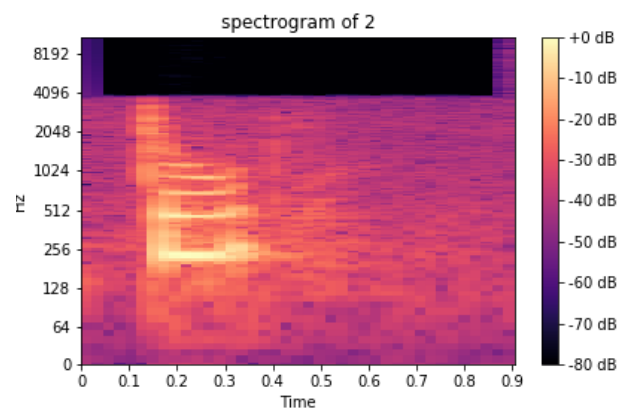
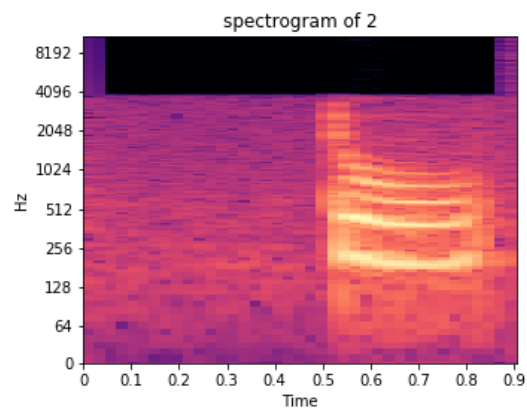
Utterance of the Word Zero



Utterance of the Word One



Utterance of the Word Two

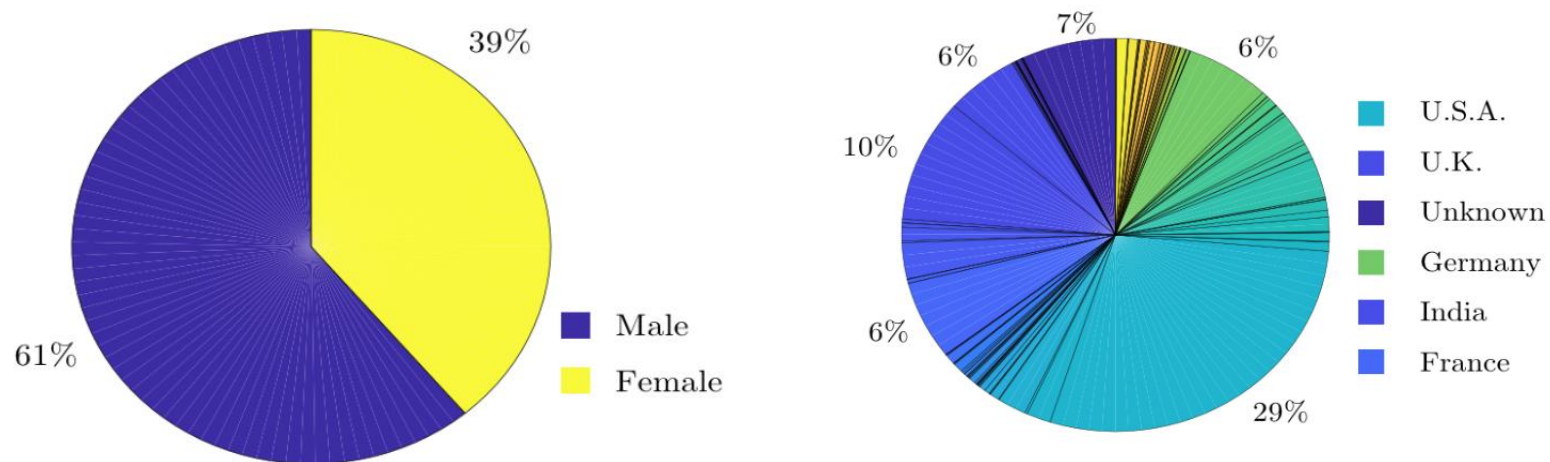


Example problem

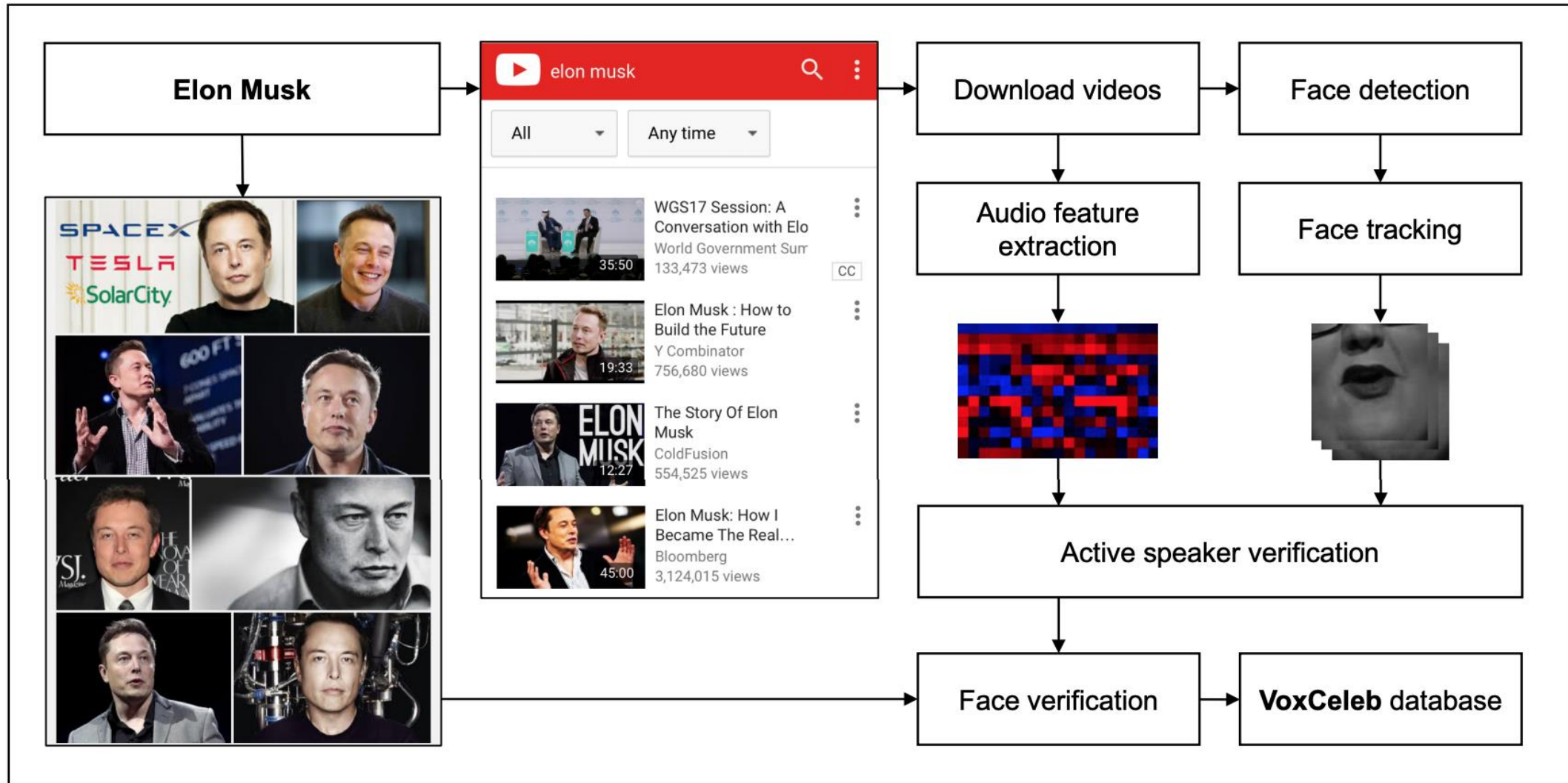


VoxCeleb2

VoxCeleb2 contains over a million utterances for 6,112 identities.



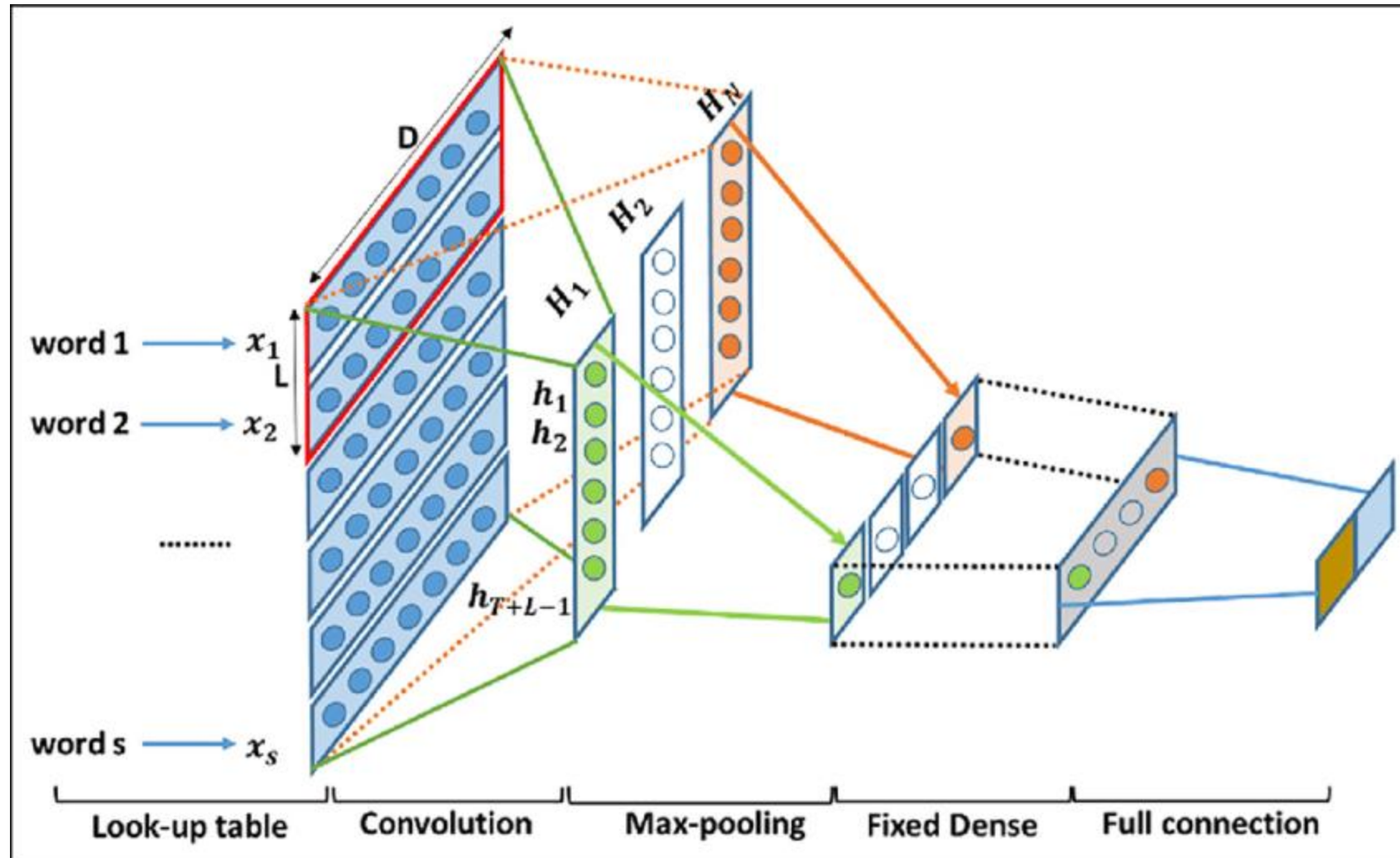
Example problem



Performance on VoxCeleb

Accuracy	Top-1 (%)	Top-5 (%)
I-vectors + SVM	49.0	56.6
I-vectors + PLDA + SVM	60.8	75.6
CNN-fc-3s no var. norm.	63.5	80.3
CNN-fc-3s	72.4	87.4
CNN	80.5	92.1

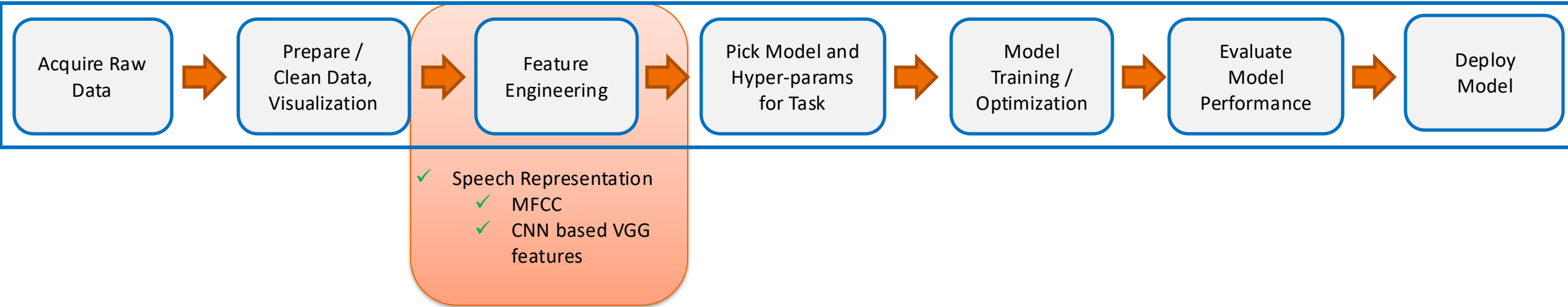
Neural Networks + word2vec for text



Summary

- Data driven features are now effective for many data.
 - “Learn from some one else’s data”.
 - “Refine to your problem” (more later)
- Many recognition/classification tasks in the image and speech space are reachable.

Summary



Thanks!!

Questions?